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Review

A Review of Internet of Things Approaches for Vehicle Accident Detection and Emergency Notification

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Abstract

The inspiration behind this specific research is based on addressing the growing need to improve road safety via the application of the Internet of Things (IoT) system. Although several investigations have discovered the possibility of IoT-based accident recognition, recent research remains fragmented, usually concentrating on outdated science or specific use cases. This study aims to fill that gap by carefully examining and conducting a comparative analysis of 101 peer-reviewed articles published between 2008 and 2025, with a focus on IoT systems for accident recognition techniques. The review categorizes approaches depending on the sensor used, incorporation frameworks, and recognition techniques. The study examines numerous sensors, such as Global System for Mobile Communications/Global Positioning System (GSM/GPS), accelerometers, vibration, and many other superior sensors. The research shows the constraints and advantages of existing techniques, concentrating on the significance of multi-sensor utilization in enhancing recognition precision and dependability. Findings indicate that, although substantial improvements have been made in the use of IoT-based systems for accident recognition, problems such as substantial implementation costs, weather conditions, and data precision issues persist. Moreover, the research acknowledges deficiencies in standardization, as well as the requirement for strong communication systems to enhance the responsiveness of emergency services. As a result, the study suggests a plan for upcoming developments, concentrating on the incorporation of IoT-enabled infrastructure, sensor fusion approaches, and artificial intelligence. This study improves knowledge by offering an extensive viewpoint on IoT-based accident recognition, providing insights for upcoming research, and suggesting policies to facilitate implementation, eventually enhancing worldwide road safety.

Keywords: road safety; Internet of Things (IoT); accident detection; sensor

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1. Introduction

1.1. Background

Annually, over 1.19 million individuals lose their lives due to road traffic accidents. Between 20 and 50 million more individuals have minor injuries, with many cases resulting in disabilities. Accidental injuries result in significant economic detriment to people, relatives, and governments collectively. The damages stem from treatment expenses and diminished production for those who pass away or are incapacitated due to injuries sustained, as well as for relatives who must take leave from work or school to attend to the wounded. Highway traffic accidents incur costs equivalent to 3% of a nation's gross domestic product. [1]. In the United States, automobile collisions are regarded as a primary

cause of mortality. In 2020, over 40,000 individuals perished in automobile accidents, while around 2.1 million sought treatments in emergency units because of such incidents. The projected cost is USD 430 billion, including medical expenses, diminished quality of life, and fatalities [2,3].

In recent years, the Internet of Things (IoT) has emerged as a significant technology of the twenty-first century. Currently, people are able to link daily items, e.g., kitchen devices, automobiles, industry, and so on, to the internet through embedded systems, and seamless connection is feasible among individuals, processes, and objects. Through low-cost processing, big data analysis, the cloud, and smartphone systems, physical items are able to share as well as gather information with minimal individual involvement. In this particular hyperconnected globe, digital platforms are able to document, track, and adjust every connection among linked items [4].

IoT has several different applications that cover practically all parts of daily life. It addresses several fields, for example, transport, farming, smart cities, supply chain, industry, healthcare, energy, environment, and so on, as shown in Figure 1. For healthcare, IoT can be utilized for patient tracking, medical cold storage, fall recognition, and dental applications. For an intelligent environment, it can be used for temperature tracking, disaster avoidance, water quality checking, wildlife protection, forest fire recognition, and air quality tracking. The function of IoT in industrial sectors generally includes monitoring for explosive and hazardous gases, checking gas, oil, and water quantities, performing repairs and maintenance, controlling vehicle fleets, monitoring indoor air quality, recognizing ozone, tracking temperatures, and initiating recovery actions. In farming, it can be utilized for air quality checking, animal, greenhouse, and field tracking, water and soil management, and pest management. In the automotive field, it can be used in numerous areas, for example, automobile servicing, automobile monitoring, automated toll and fine payments, accident recognition, fire recognition, entertainment with connected vehicles, velocity tracking, and motorist recognition [5].

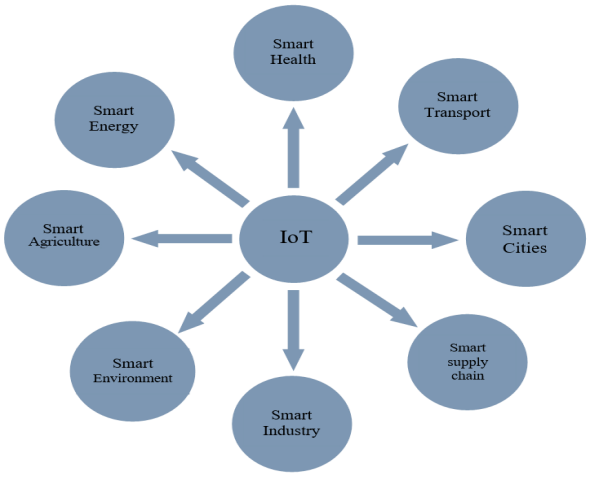

Figure 1. Application of IoT.

Since IoT is actually an interconnection of a wide range of intelligent and embedded units, for example, sensors, mobile phones, computers, and embedded processors, along with the modern internet, it can be utilized to reduce the vehicle crash rescue period. Several investigations, such as Bhatti et al. [6], Kumar et al. [7], Parveen et al. [8], Biswal et al. [9], Zhen et al. [10], and Uma & Eswari [11], confined their research to enhancing the precision of vehicle crash recognition, determining the intensity of highway crashes, or reducing the rescue time post-incident. Some other researchers, such as Bhakat et al. [12], Balfaqih et al. [13], Pathik et al. [14], Geetha et al. [15], and Zavantis et al. [16] worked

on IoT systems to detect different types of accidents, such as head-on collisions, rear-end collisions, rollovers, sideswipes, and so on. Details regarding the form of vehicle crash sustained give valuable data about the injuries sustained in automobiles and the level of damage sustained by the sufferers. Therefore, the alarming occurrence of vehicle crashes significantly enhances the capability of the EMS to give the sufferers the correct form of medical aid.

1.2. IOT Challenges

Implementing the IoT in vehicle crash detection and alert systems raises several issues that must be addressed to ensure performance and dependability.

Data Privacy and Security: IoT platforms gather and transfer sensitive details, for example, automobile data and individual information. Guaranteeing the privacy and integrity of this particular information is vital, as breaches can result in illegal accessibility and improper use. Establishing strong encryption techniques as well as safe connection methods is important to keep user privacy [17].

Interoperability: The IoT platforms include various equipment as well as systems from different companies, generally working on diverse specifications. Achieving seamless connection among these types of heterogeneous platforms is difficult. Creating general specifications and standards is important to allow interoperability as well as guarantee cohesive function throughout various equipment [14].

Energy Efficiency: Several IoT products, particularly sensors utilized in vehicle crash recognition, are generally implemented in environments where frequent servicing is challenging. These types of products frequently depend on battery power, making power performance crucial. Establishing optimized software functions as well as low-power hardware is generally essential to extending product lifetime and keeping the platform dependable [18].

Network Connectivity and Latency: Vehicle crash recognition platforms need live information to be transmitted quickly to notify emergency providers. Nevertheless, network coverage may be inconsistent, and latency concerns can occur, especially in distant locations. Guaranteeing a dependable and low-latency connection is undoubtedly essential to keep the performance of these kinds of platforms [19].

Addressing these particular issues requires a comprehensive strategy that includes energy-efficient design, standardization efforts, enhancements in cybersecurity, as well as the advancement of strong connection networks. Collaborative attempts among policymakers, industry stakeholders, and scientists are generally important to overcome these types of challenges as well as to entirely understand the potential of IoT in improving vehicle crash detection and alert systems.

1.3. Motivation Behind This Study

The quick technical enhancements of the twenty-first century have substantially influenced different areas, such as the transport sector. In spite of these kinds of innovations, highway traffic crashes continue to be a crucial worldwide problem, resulting in large numbers of fatalities, serious injuries, as well as significant social and economic burdens annually. Based on worldwide information, delayed reactions to vehicle crashes and the failure to alert emergency authorities are substantial contributors to the loss of life in such occurrences. Traditional techniques of crash recognition and communication, which are frequently dependent on outdated platforms or manual intervention, fail to give the quickness and precision required for well-timed involvement.

The IoT-based systems have appeared as a transformative option with great potential to revolutionize vehicle crash detection and alert systems. It allows seamless connection

among devices, enabling live tracking, automation, and information exchange. By combining enhanced sensors, for example, GSM, GPS, gyroscopes, accelerometers, and ultrasonic, IoT-based platforms can identify vehicle crashes with excessive accuracy and exchange important details with authorities, relatives, or emergency responders in seconds. This ability is essential in decreasing the time period between a crash and the arrival of assistance, considerably enhancing survival rates.

The inspiration behind this particular research lies in addressing these continual issues and discovering IoT's role in improving highway safety. An extensive review of IoT applications in vehicle crash recognition and notice systems is important, as the current literature is fragmented, usually concentrating on isolated systems or restricted utilization conditions. This research attempts to fill this particular gap by delivering an in-depth evaluation of existing platforms, discovering technical and functional restrictions, and suggesting a plan for upcoming enhancements. This research can encourage experts, industry stakeholders, and policymakers to establish and adopt strong IoT solutions, which can decrease death rates and enhance the total performance of emergency reaction platforms and traffic management.

This research is arranged as follows. Section 2 provides a thorough review of existing literature on IoT accident detection and notification, spanning 24 years from 2000. Section 3 presents a comparative analysis and discussion of the works described in Section 2, which is followed by policy suggestions and limitations of this study in Section 4. Section 5 provides a conclusion and suggestions for future work as a summary of the current research.

2. Literature Review

2.1. Early Contributions (2008–2010)

An innovative technique for accident recognition was provided by Li et al. [20]; it utilizes an in-car terminal that includes a GSM/GPS component and a control module, as well as several optional elements, such as a CPS module and airbag sensors. A motorist or automobile beginning an alert report would be instantly located by CPS, GPS, or both, accompanied by a zoom-in using Closed-Circuit Television (CCTV) to validate the incident.

Thompson et al. [21] presented three contributions to the research on utilizing mobile-based crash recognition techniques. They explained solutions to key issues related to finding vehicle crashes, for example, avoiding false alerts by using smartphone context information as well as polling onboard sensors to identify enormous accelerations. Thompson et al. [21] noted that mobile-based crash recognition can decrease total traffic jams as well as enhance the readiness of emergency activities.

Chae and Yoshida [22] explained the use of Radio Frequency Identification (RFID) systems to avoid crashes with serious devices, for example, hydraulic cranes and excavators. A system design is suggested that utilizes RFID systems, which provides a function through a defined role. A prototype has been created utilizing a kind of RFID tag to facilitate the functions of a support system.

2.2. Mid-Term Advances (2011–2015)

White et al. [23] explained how mobile phones, for example the Android and iPhone systems, can instantly identify vehicle crashes utilizing acoustic information and accelerometers, quickly alert a central emergency dispatch server after a crash, as well as determine situational consciousness via VOIP communication channels, GPS, photographs, and crash information recording. Zaldivar et al. [24] suggested an application based on the Android system, which monitors the automobile via a system named On-Board Diagnostics (OBD) and is capable of recognizing crashes. They suggested that a mobile app calculates the G-force encountered by people in vehicles in case of a frontal accident, which is utilized

together with airbag triggers to recognize crashes. This system responds to positive recognition by transmitting information regarding the crash via Short Message Service (SMS) or e-mail to pre-defined destinations, quickly accompanied by a programmed phone call to the Emergency Medical Service (EMS).

A system named e-NOTIFY was actually introduced by Fogue et al. [25], enabling quick recognition of vehicle crashes, enhancing the support to damaged passengers through decreasing the response period of EMS via the effective communication of specific details regarding the crash utilizing a combination of Vehicle-to-Infrastructure (V2I) and Vehicle-to-Vehicle (V2V) communications. The suggested technique requires installing an onboard unit (OBU) in the automobiles, which is responsible for detecting crashes and informing an external Control Unit (CU), which will then calculate the severity of the crash and notify the appropriate EMS about the accident. Amin et al. [26] suggested using the functionality of a GPS device to screen the velocity of an automobile as well as recognize a crash based on monitored velocity, then transmit the crash area to a specific center. The GPS module will monitor the velocity of a car as well as compare it with the prior velocity every second via a microcontroller device. If the vehicle speed drops below a certain level, it can be presumed that a crash has happened. Then, the platform can transmit the crash location obtained through the GPS device, along with the period and velocity, using the GSM module.

Wireless black boxes utilizing a GPS module and micro-electromechanical system (MEMS) accelerometer were actually designed by Watthanawisuth et al. [27] for vehicle crash monitoring. The system comprises cooperative elements, including a microcontroller device, an accelerometer, and a GSM/GPS module. In the event of a crash, this wireless module can transmit a smartphone message showing the location of the automobile as determined by the GPS module to EMS, the closest medical center, and a designated family member. The threshold algorithm, as well as the velocity of the motorbike, are generally utilized to identify a crash or fall in real time. The test outcomes demonstrate that it can identify normal rides, non-linear falls, and linear falls with excessive precision. Amin et al. [28] suggested a system able to identify a crash from the map-matched location of an automobile by using the GPS velocity information and a matching algorithm, as well as transmit the crash location to an alarm support center. Anytime the velocity drops below the securely determined limit velocity, the system will create a crash circumstance. It can examine the automobile's location through the map matching module as well as create a crash circumstance if the automobile is actually discovered outside the highway network.

Sherif et al. [29] developed a real-time traffic accident detection technique utilizing RFID and wireless sensor network systems. Sensors mounted in an automobile recognize the crash area, the automobile's velocity just before the crash, as well as the range of passengers in the automobile. Then, the sensors transmit a notify signal to a monitoring station. Nazir et al. [30] offered a solution for an incident recognition and avoidance system designed to save human lives. It enables smart recognition of a vehicle crash at any location and reports about the crash to predetermined phone numbers. The system consists of a GSM module, a microcontroller, SONAR ranging modules, GPS, a vibration sensor, and an alert system. When the distance is quite short between the automobile and the barrier, the alert will be "ON" as an indicator to shift automobile onto another path which is less dangerous; however, when the automobile experiences a crash in spite of the alert, vibration sensor will instantly identify the signal after which the microcontroller uses GSM to transmit the alarm message, consisting of the location, to predetermined phone numbers which can be reserved for a rescue team.

Fernandes et al. [31] introduced a mobile application for vehicle crash recognition integrated with multimodal alarm distribution, utilizing either IEEE 802.11p or eCall. The suggested car crash recognition algorithm utilized some sensors as mentioned in

Table 1. The mentioned application is actually utilized as a human–machine interface, allowing motorists to configure the application, receive highway hazard alerts from other automobiles in the vicinity, as well as terminate countdown processes upon incorrect crash recognition. Aloul et al. [32] introduced a system that utilizes mobile phones to identify and report vehicle crashes in a timely manner. Information is actually constantly gathered through the mobile phone's accelerometer and examined utilizing the Hidden Markov approach as well as dynamic time warping to figure out the degree of the crash, decrease false alarms, and inform initial responders of the crash location and the owner's healthcare data.

2.3. Recent Innovations (2016–2024)

Sankar et al. [33] suggested an extensive solution for ambulance management and car crash recognition. When the in-vehicle incident recognition system reports a crash, the principal server instantly dispatches the closest ambulance to the location of the car crash. This mobile application, utilized by the ambulance driver, facilitates the ambulance's safe and rapid arrival at the accident site. Sharma et al. [34] suggested an automatic car crash recognition system based on a mobile phone. Their algorithm consistently infers car crashes depending on the information gathered via numerous mobile phone detectors, despite their range limitations and sensitivity. This system offers a less expensive option to the expensive car crash recognition systems commonly found in luxury automobiles.

Ibrahim et al. [35] designed a system for discovering automobile accidents as well as rollover accidents that consists of three key phases: information collection, information filtration and evaluation, and notification. Information collection using motion sensors, gyros, and accelerometers. An algorithm for gathering information utilizing sensors, filtering, and analyzing data was established to identify accidents and rollovers. Smolka and Skublewska-Paszkowska [36] utilized a mobile phone, which not only allows for the instant transmission of data about the accident but also enables automatic voice connection. The suggested procedure can also serve as a "black box," saving data on a moving automobile before the accident. In the suggested system, it was feasible to create an emergency phone number. Rather than EMS, family members can be informed and, if necessary, they can inform EMS.

An IoT-based car crash recognition and rescue information system was created by Sany and Riyadh [37] to discover car crashes and transmit the location data of the crash area to the automobile owner, the closest medical center, and the police station through a web service. The communication between the web server and components is set up through a General Packet Radio Service/Global System for Mobile Communications (GPRS/GSM) module, and the area is tracked by utilizing the GPS module. The crash is actually recognized via buzzer and vibration sensors. Celesti et al. [38] described a feasible option for directly incorporating smartphone sensors into private and public transportation, as well as volunteer automobiles. In this particular circumstance, quick live processing of large amounts of traffic information is essential to avoid a car crash. Celesti et al. [38] mentioned an IoT Cloud platform for traffic monitoring as well as alarm communication utilizing MongoDB and Open GTS. Their IoT Cloud platform, apart from private drivers, is really beneficial for drivers of critical rescue automobiles, for example, EMS.

Mohammed and Kamsani et al. [39] suggested offering medical assistance as well as first aid kits prior to the arrival of EMS. This innovation is both a software and a hardware system, which works together for car crash recognition, reporting, and a medical assistance program. It was created to deliver an automatic crash reporting system and also to save human lives. Zualkernan et al. [40] developed a smartphone application and installed it inside a vehicle where it read the phone's gyroscope and accelerometer every 5 milliseconds.

This information is entered into an artificial intelligence (AI) classifier model, which can determine if a car crash has occurred. The software instantly transmits a notice to local EMS, police, and emergency phone numbers regarding a car crash. This SMS notice consists of the location and type of the crash.

Shaik et al. [41] explained the feasibility of installing an automobile with systems that can recognize a crash and instantly notify EMS utilizing IoT. A signal from a GPS and an accelerometer is instantly delivered to the cloud, and then a notification message can be transferred to anyone who is subscribed to that vehicle. The message can indicate the location and severity of the crash. The EMS can utilize GPS coordinates to quickly reach the field. Khaliq et al. [42] introduced a TestBed, which is developed and integrated with the prototype software to offer an option for road traffic safety utilizing Vehicular Ad hoc Network (VANET) and IOT. The software finds and analyzes the intensity of crashes with the assistance of an OBU, used in automobiles. When a car crash occurs, it triggers an alert in the control room, where a server decides the location of the accident as well as the closest medical center to receive fast healthcare assistance.

An automatic alert system for automobile crashes is presented by Fernandez et al. [43]. This particular system is able to recognize crashes considerably faster and transmit simple data to medical centers within a few seconds, including location, angle, and the time at which the automobile crash occurred. The information can be delivered via the GSM module, and the crash area can be recognized via the GPS module. An intelligent crash recognition and control system regarding intersecting roads in highway networks has been suggested by Hadi et al. [44]. This innovation utilizes a microcontroller, which is not required in every automobile, but is needed in an OBU. The system is able to identify the area of the impacted automobiles as well as to quickly determine the appropriate actions to eliminate the worst circumstances. It is taken into account as a digital platform to instantly identify the inevitable conditions on the highway.

To detect and prevent accidents, Al Wadhahi et al. [45] developed a system utilizing Arduino Uno and infrared (IR) sensor technology. The recognition step is performed utilizing IR sensors that can identify and alarm the individuals through delivering SMS utilizing a GSM sensor, which includes predetermined phone numbers and the crash area location, utilizing a GPS sensor. The second stage, crash avoidance, is performed utilizing IR sensors by alerting the motorist about nearby automobiles when the distance between them is further than the threshold. Dhanya et al. [46] developed a system and a sensor that can identify the severity of an accident and transmit this information to the controller. As soon as the threshold is crossed, the sensor can transmit the notification information to the control station or recovery group through the ZigBee sensor. Therefore, the recovery group can instantly track the area, and the required measures can be taken.

A new strategy for recognizing a highway crash utilizing a low-cost ultrasonic module is actually suggested by Khalil et al. [47]. Even though this method is adequate for identifying a crash, every method has its limitations. To assess the efficiency of a suggested model, various simulations are carried out utilizing MATLAB to discover the elements that impact the recognition of a crash utilizing an ultrasonic module. Dar et al. [48] leveraged the benefits of fog computing to create a fog-based crash recognition as well as emergency reaction that utilizes Android gadget sensors to identify an emergency incident and crash area, and generate an action strategy to manage the emergency in a timely manner. Dar et al. [48] examined a strategy using iFogSim, an open-source tool for simulating fog systems alongside Cloud and IoT.

Vatti et al. [49] tried to create a vehicle crash recognition and connection model that can notify the family, the closest medical center, as well as police near the crash area. Likewise, if the automobile does not crash but tilts or topples by any significant angle, the platform

can identify the crash from the gyroscopic module. The heart rate module can identify the heartbeat of the motorist, the present area can be gathered through the GPS sensor, and the SMS module can be used to deliver messages to urgent contacts. Fanca et al. [50] developed a platform for discovering and reporting incidents, and decreasing the time period between a vehicle crash and delivering the initial urgent alarm to the specific center. This approach provides a solution utilizing a mobile phone for discovering and avoiding crashes, such as falls.

A system was developed by Rakhonde et al. [51] to identify vehicle crashes in real-time as well as reduce the reaction time for medical assistance. For vehicle crash prevention, tire pressure is actually assessed, whereas in crash recognition, it is applied with the assistance of a Node MCU. MQ7 is utilized to monitor environmental pollution. The suggested platform is valuable in decreasing the rate of vehicle crashes, as well as in pollution monitoring, which can assist in understanding environmental conditions. Dias et al. [52] established a car monitoring and crash notification platform to prevent delays in medical aid to victims. The automobile monitoring platform offers security to most automobiles by constantly monitoring them, whereas crash notification rescues individuals involved in a crash by discovering crashes with the assistance of several sensors. The platform offers enhanced security to most automobiles, as well as avoiding death because of delay treatment through the setup of GSM, GPRS, and GPS modules.

Nanda et al. [53] suggested a procedure that can efficiently assist in avoiding accidents, as well as if such circumstances happen subsequently, how it identifies and notifies the involved individuals and authorities, so that the circumstance can be addressed promptly. This particular procedure detects crashes through accelerometers and vibration modules. For recognition, they utilized GSM and GPS modules that locate the crash, as well as notify the person's close ones and the nearby medical center via a short message. Mankar and Tasgoankar [54] created the ALDSTM platform that can instantly provide the crash area coordinates to the rescue group, enabling them to provide healthcare assistance to the victim as early as feasible to save lives. In addition, it can assist the medical group by monitoring the patient's heartbeat in the vehicle prior to the arrival at the accident location by utilizing a pulse module. In this regard, GPS and GSM modules are utilized to obtain area coordinates and transfer message to the medical center, while accelerometer sensors are utilized to identify the crash.

Taj et al. [55] introduced an Automatic Crash Recognition and Human Recovery Platform, which can identify crashes as well as save people by updating the rescue group. GPS, GSM, and vibration modules are generally utilized in this platform. After obtaining the location of a crash, the platform transfers information to a police station and a close rescue group through the GSM sensor. A rescue group can quickly track the area of the automobile on Google Earth to assist injured people. An Android program established by Dar et al. [56] used mobile phone sensor modules for accident recognition. When a car crash is actually recognized, the first step is to locate a nearby medical center using the GPS module. The emergency office of the medical center is informed about the car crash, which directs an EMS to the crash area. Furthermore, the predetermined relatives of the victim will be informed about the crash.

Devi and Pamila [57] offered a modern option by establishing a crash short alarm information system utilizing an Android mobile phone program which can be utilized in the crash area. The program utilizes a GPS module for the area, as well as transmits an alarm and a notice of a crash. The created vehicle crash information is actually supported by the registered users who observe the crash to enhance the status of the message. Sarker et al. [58] established a smart crash recognition, area tracking, and notification platform that instantly detects a crash whenever it happens. The GPS module discovers the precise

area of the crash, and the GSM component transmits a notice, such as a link to the area on Google Maps, to the closest police station and medical center, enabling them to verify the location. They can also discover the quickest route to the crash location and attempt to accelerate the rescue procedure.

An accessible low-cost automatic crash recognition system was suggested by Hassan et al. [59], which includes instantly finding the accident and transmitting a notice via SMS to the EMS, as well as the involved relatives, with an accurate area. The platform is actually dependent on the Arduino microcontroller, which incorporates accelerometer and vibration modules commonly used in automobiles. The software section, on the other hand, is an Android program installed on the user's smartphone. Shankarpure and Abin [60] suggested a new strategy to notify the close hospitals about the crash, enabling them to offer instant medical-related aid. The platform can utilize an accelerometer sensor in smartphones to perceive the tilt of the automobile. Therefore, the platform will notify the center and deliver the data via the smartphone application. The Android program on the smartphone will transmit short messages to the closest hospital, automobile owner, and relatives. The program can show the precise area of the crash to the EMS.

Chang et al. [61] established a technique for discovering high-speed head-on as well as single automobile accidents, examining the circumstances, and sending an alert. Chang et al. [61] suggested a method known as DeepCrash, which uses deep learning-based Internet of Vehicles (IoV). This approach consists of an in-vehicle infotainment telematics system equipped with an automobile self-collision recognition module, a cloud-based management system, a cloud-based deep learning (DL) server, and a front-facing camera. When a single-vehicle or head-on accident is recognized, crash recognition details are actually submitted to the cloud-based platform for self-collision automobile crash identification, and an associated urgent notice can be offered. Ashokkumar et al. [62] developed an accident detection system using a vibration sensor, alcohol detection, and an eye blink sensor. If a vehicle rolls over, the vibration module can instantly identify the signal. Whenever an automobile is involved in an incident, the module can locate the signal, transmit it to the server, and then notify the family or a rescue group.

To offer a fast and suitable connection between a vehicle crash and appropriate medical assistance, Choudhury et al. [63] developed a system utilizing an Arduino Uno microcontroller as the central part of the gadget. In addition, they utilized an accelerometer module to identify the crash, a heartbeat sensor for tracking the heartbeat of the motorist, a GPS sensor to identify the area of the crash as well as a GSM sensor to transmit an SMS to the closest medical center, closest police station, and a family member of the motorist using a google maps connection to the area and heartbeat of the motorist. Patil et al. [64] utilized IoT to identify vehicle crashes and notify the relevant parties of the crash severity level for urgent situations, as well as to determine the vehicle's health with particular details. Mobile phones, GPS and GSM modules, vehicular ad hoc networks, as well as smartphone software are generally utilized in crash recognition. Data can be delivered to the closest medical center and police station by utilizing a system that precisely locates the vehicle crash.

The main inspiration behind the system established by Gowri et al. [65] was to notify relatives of the injured person about the accident, as well as to gradually assess the motorist's condition in terms of drowsiness or weariness. The vibration module finds the incident, and the heartbeat module investigates the heartbeat of the motorist. Whenever the heartbeat degree is actually high and vibration occurs, it can be anticipated that a crash has occurred, as information revealing the driver's heartbeat and location can be delivered to their family members. The eye flicker module is actually installed in a wearable glass. Whenever it identifies that the eyes are generally closed, the alert bell begins to sound. An IoT-based platform is suggested by Kader et al. [66] to ensure continuous monitoring of

an automobile's details, such as car velocity, rolling of the automobile, and range of harsh brakes that can be used to evaluate the level of vehicle travel. Whenever the automobile's velocity is greater than the speed limit, the platform can notify the motorist by warning on the buzzer. Furthermore, the platform has a switch, and if this particular switch is pushed, the platform can deliver a short message immediately to correct authority with the area details.

Parteki et al. [67] suggested a platform that can offer the quickest possible assistance to casualties. The platform in the automobile would first identify the crash utilizing the vibration module, identify the area of the crash utilizing the GPS module, and then deliver alert information to EMS or a nearby medical center utilizing a GSM device. As the EMS receives the message, it can find the quickest road to the accident area by switching on the RF antenna that receives signals from all signalized intersections between the crash area and the EMS location. The suggested strategy by Bhatti et al. [6] focused on leveraging the advanced features of mobile phones to establish a low-cost solution regarding an improved transport network. A personalized Android program is established to collect data concerning velocity, area, pressure, noise, and gravitational force. Moreover, a navigation platform is actually established to report the accident to the closest medical center.

A vehicle crash avoidance and EWS system, utilizing wireless Bluetooth systems was suggested by Wang et al. [68], consisting of three components: central control, ranging, and an Android smartphone system. The results demonstrate that when the automobile is traveling at low speeds, the ranging error is between 0 and 10 cm, which is comparable to radar ranging error. The acoustic-optical alert and real-time distance monitoring can better remind the driver to control the vehicle's following distance to prevent rear-end collisions at low speeds, as well as to avoid rear accidents while reversing and side accidents caused by blind spots. Kashevnik et al. [69] introduced a technique and smartphone program for motorist tracking and evaluation depending on recognized dangerous driving behavior, utilizing the individual mobile phone's cameras as well as sensors such as a microphone, GPS, gyroscope, and accelerometer. The technique works by following motorist's online hazardous states, such as sleepiness and distraction. Kumar et al. [70] showed an IoT platform that utilizes the built-in modules of a mobile phone to report highway crashes precisely. The National Highway Traffic Safety Administration utilizes information to test and train its crash forecasting models. The forecasting efficiency was discovered to be extremely precise, with a MAPE of 3.49% and 2.34%, respectively, regarding the information utilized.

Kumar et al. [7,71] displayed an IoT-based automotive crash recognition and classification platform that utilizes the fusion of mobile phone modules as well as linked detectors not only to identify but also to review the types of crashes, such as falls, rollovers, and non-crashes. This particular strategy enhances the effectiveness of savings, for example, EMS, towing services, and fire stations. This system decreases the time taken regarding communication after crash recognition by using machine learning (ML) algorithms directly on the mobile phone itself. Karmokar et al. [72] suggested an automated IoT-based efficient crash recognition platform utilizing GSM and Wi-Fi modules. Instantly after an accident, the dataset is actually delivered to the web server, and an immediate short message can be sent to the victim's family and related authorities, for example, EMS, the closest police station, and the traffic control room.

Rishi et al. [73] facilitated the real-time pursuit of a vehicle and tried to reduce the probability of fatalities by delaying the arrival of EMS through notifying the worried individuals about the automobile accident. To establish this type of platform that can alert the worried individuals, accelerometer, GSM, and GPS sensors are actually interfaced with an Arduino Uno microcontroller. The accelerometer module finds a crash by measuring a

change in the predetermined magnitude of the automobile's orientation, and then transmits the area via a GPS sensor to a registered SIM card through GSM to the motorist or travelers. Ajao et al. [74] established an embedded in-vehicle crash recognition as well as a barrier recognition notice platform. A radar sensor is actually suggested for barrier recognition on the highway prior to an accident happening. Furthermore, a distance–time-based variables algorithm was established for recognizing crash area coordinates utilizing the Haversine method. An IoT system was used by combining wireless connection components and a GPS/GSM module, as well as an accelerometer/gyroscope sensor module.

Rana et al. [75] presented a faster emergency reaction using a smartphone program that transmits the live coordination of the crash area utilizing a GPS sensor. The two distinct smartphone programs within this particular platform comprise a driver mode for synchronizing their account with the platform, a pedestrian section for delivering videos and photos of the crash area, and an EMS webpage, which receives all notifications and details of all crashes happening. Kapilan et al. [76] develop an IoT system to identify crashes and report them to the area coordination of the accident via a predetermined phone number using a shock sensor, an accelerometer, and a GPS/GSM module. With this particular system, instant behavior can be considered in the event of a crash. It can assist the EMS in arriving at the crash site soon and save individual lives.

Aung and Thein [77] developed a system to identify road crash injuries by utilizing information obtained from several sensors. This system is supposed to utilize the mobile phone's built-in high-tech sensors, that are linked using Geofence, GPS, and GIS technologies. Since this particular strategy is applied in the user-friendly edition, the individual can basically respond by pushing the confirmation message to indicate whether they require the help of the rescue group or not. Habib et al. [78] tried to establish a system to decrease the rates of injury and fatality among individuals involved in household fires as well as highway crashes. The platform is thoroughly automated to deliver urgent notices to nearby medical centers, fire stations, or police stations. This can be carried out through a combination of a system setup inside the automobile, as well as a remote server platform that responds to the device accordingly.

Kashevnik et al. [79] introduced a strategy that uses mobile phone sensors for discovering hazardous states related to a motorist in an automobile. They utilized a mobile phone installed on an automobile windshield, directed towards the motorist's face, and monitored through the front-facing camera. Utilizing details from camera video frames and other sensors, they identified high pulse rates, hazardous conditions, distraction, violent driving, and drowsiness, which can result in highway crashes. The cloud service offers reviews on motorist travels and data to developers. Bhakat et al. [12] utilized IoT, ML methods, and image processing to determine highway crashes precisely. The sensors, such as camera, gyroscope, accelerometer, etc., offer information to a microprocessor that fits the sensor information with the ML methods as well as decides if there is a crash or not, and in case of an accident, the system transmits the associated metrics to the server via the web. In addition, when the information reaches the server, it establishes the closest police stations and medical center using the GPS module and transmits a message to them as well as to the predetermined telephone number through the end user.

Rehman et al. [80] proposed the development of a crash recognition platform for motorbikes, which alerts the rider of a damaged motorbike to their precise location, enabling timely healthcare to be provided. The suggested platform actually depends on a tilt module, which computes the inclination of the motorbike and sends a notice to the worried individuals via SMS and GPRS through an online machine utilizing a GSM component. Chikaka & Longe [81] released an automated automobile crash recognition system along with a notification platform that utilizes an accelerometer module to identify automobile

crashes and transmit the GPS area of the crash to the medical center and concerned relatives. The platform finds a crash when the arranged threshold magnitudes are exceeded, and notifies the authorities and predetermined relatives within 4 s.

Yellamma et al. [82] suggested a platform for delivering messages to individual's relatives using GSM and GPS sensors. An LCD is utilized to display messages and the location of the crash. Babalola & Olokun [83] concentrated on creating an automobile crash recognition system utilizing GPS/GSM sensors. With signals from the accelerometer sensor, a serious crash can be identified. Whenever an automobile is involved in an accident, the vibration module can identify the signal, or if it rolls over, a MEMS device can identify the related signal. The microcontroller transmits the SMS via the GPS/GSM sensors to the closest police station, allowing the group to easily track the area using the GPS module. Narayanan et al. [84] proposed an IoT-based intelligent crash recognition system, along with an insurance claiming platform, to rapidly detect the crash area and velocity of an automobile in a crash, utilizing GPS/GSM, vibration, and Wi-Fi modules. This particular platform not only identifies a crash but also transfers the crash information to the authorities via a cloud server. This platform notifies the insurance organization about the crash, which can be helpful for the person in speeding up the insurance claim process.

Chaithanya et al. [85] introduced a platform that, once a crash is detected, triggers a buzzer. If the driver does not respond to the buzzer within a fixed time period, the GPS sensor updates the location via Wi-Fi, along with eye blink readings, heartbeat data, and information about which side the victim has fallen. Immediately after the accident, a GSM sensor transmits the location and cloud link address to the mobile phone connected to the platform. Instantly after an SMS, a notification call can be received through predetermined emergency contacts, and the closest hospital can be informed about the victim's location. Alkinani et al. [86] suggested a crash recognition platform for an intelligent city, which is suitable for any automobile and more affordable. They established an android program which gathers information associated with audio, pressure, gravitational force, velocity, and area of the crash through the mobile phone. The magnitude of velocity helps enhance the precision of crash recognition. A navigation platform was developed to notify the family, the closest medical center, and the police station.

Kumar et al. [87] suggested a simulation model that finds crashes utilizing several modules. The circuit, incorporating the module, is integrated with an Arduino microcontroller created and simulated using the Proteus Simulation Program. Whenever the crash is actually recognized, an SMS can be delivered to relatives via the application. Virtual serial port emulator software links the Proteus Simulation Program as well as the IoT system. The Blynk IoT System is actually utilized to design the prototype of the Android program that identifies crashes. Sumathy et al. [88] created an automobile crash notifying and discovery platform, which can save people from hazardous highway crashes, particularly in remote locations where individual activities are limited. The GPS module provides precise location details and sends them via SMS to the closest police station, EMS, medical center, and the victim's relatives. The EMS can be reached instantly at the crash location by utilizing the area information, and fast medical assistance can be given to the victim.

Mahadik et al. [89] suggested a platform that consists of GPS, GSM, mobile phone programs, and VANET techniques. Their platform provides timely details of EMS concerning the area of the crash, which assists in saving valuable lives. Whenever a crash happens, this kind of sensor data will be delivered to the closest medical center and police station. Vijayaraja et al. [90] established an automobile accident recognition platform utilizing and accelerometer, and both GSM and GPS modules. The accelerometer finds the abrupt alteration in the axes of the automobile, and the GSM sensor transmits a notification message to the smartphone with the area of the crash. Through the GPS module, the location of

the crash can be pinpointed. The message also includes the velocity of the automobile in kilometers per hour.

Sampoornam et al. [91] applied ideas for the life-saving and preventive actions to avoid vehicle crashes. By utilizing GPS and GSM modules, the exact location of the vehicle crash can be transmitted to the closest EMS, relatives, and police station. The occurrence of a crash can be recognized by utilizing an accelerometer module, and the intensity of the crash can be recognized by utilizing a tilt module. The sleepiness of the motorist can be recognized by using a drowsiness sensor. All the information can be organized in a microcontroller with the assistance of sensor magnitude and transmitted to the cloud. Balfaqih et al. [13] established a platform depending on IoT for crash recognition as well as classification regarding intensity degree, and reported important details about vehicle crashes to EMS. The platform includes a GPS, a microcontroller, and a collection of detectors to identify various physical variables associated with automobile movement. The results demonstrated that the intensity of crashes generally depends on the magnitude of g-force and the occurrence of a fire.

A smartphone tracking program for EMS is suggested by Sasipriya et al. [92], who also highlight different challenges in the automobile monitoring platform. This program allows the EMS motorist to report its accessibility and area. The person who executes the emergency can reserve the EMS and log in as the end user. As soon as the patient is boarded, the EMS area can be noted and reported to the traffic police at the signalized intersections. Then, the motorist selects the closest medical center on the map that can provide timely assistance to patients. Priyath [93] suggested a new strategy for identifying real-time motorist fatigue through tracking behavioral actions, facial movements, individual physical indicators, and car variables. Face movement, for example, yawning and eye features, can be taken and examined through computer vision methods. Accordingly, this strategy was utilized to build detectors of modern wearable gadgets, such as intelligent watches, to acquire the signals. Some other sensory data, like the velocity of the automobile, heartbeat, grip pressure, and steering wheel actions, can be gathered through utilizing particular simulators and detectors. The suggested model classifies motorist sleepiness conditions in four ranges depending on input sensory information. The outcomes demonstrate that multi-sensory information, as well as the fuzzy model, offer useful factors for sleepiness recognition.

Kathiravan et al. [94] improved the emergency division's response time period to automobile crashes utilizing an "intelligent device" to record the automobile's information as well as deliver it to a third party or the car owner at standard time periods. The intelligent system was composed of a range of detectors, which are adequately protected in a black box that stays secure. The platform can help users with a range of techniques, such as offering fast support in the event of a vehicle crash, monitoring the automobile for theft, and remotely deactivating it. The system can simultaneously deliver the details to the insurance organizations immediately after the accident happens. Mehmood et al. [95] established a system that accurately delivers critical notices concerning the location of a vehicle crash, along with the vehicle's registration number, to the predetermined phone number after identifying the crash. Then, an audio call along with a saved voice is created for the predetermined phone number. Furthermore, the GPS module is utilized to obtain the location through satellite, and also the GSM module is used to send the location details of a vehicle crash.

Pathik et al. [14] suggested a smart crash recognition and recovery platform that mimics the cognitive features of the individual brain, utilizing AI and IoT systems. An IoT system can be designed to locate the crash as well as gather all details, for example, location, gravitational force, pressure, and velocity, and transmit this information to the cloud. As

soon as the crash is recognized through the DL method, all the nearest emergency services, for example, the medical center, mechanics, and police station, can be contacted. An automatic crash detection and alert system was applied by the Tippannavar et al. [96] with Raspberry Pi microcontroller for livestreaming the video in a secured website to identify the crash is happening, fuel tracking, area monitoring utilizing GPS module, bump sensors, alcohol detector, theft recognition, and accelerometer module to identify roll-off. To drive the vehicle over long ranges as well as allow encrypted communication among the receiver and transmitter, the vehicle's motions are managed utilizing an Arduino microcontroller interfaced with an RF drone controller.

A platform that utilized an accident detection system using IoT systems was proposed by Oguntimilehin et al. [97]. The platform comprises a hardware setup in an automobile as well as a web program for EMS action. The vibration module can detect a vehicle crash, and the microcontroller determines the orientation of the automobile using the tilt module, while also receiving the automobile's location through the GPS component. The microcontroller can send details regarding the crash to the web program and the nearest medical center. On a crash recognition platform developed by Selvi et al. [98], crashes can be recognized instantly using various detectors. This enables notification to EMS and numerous other emergency providers via a Smartphone Program, along with the vehicle's location, provided by a GPS module.

Tamilselvan et al. [99] offered faster and better EMS for automobile crashes. This evaluation seeks to save the individual who has been involved in a vehicle crash. Accordingly, it will deliver information, including GPS areas, to the closest hospital to notify them and encourage them to arrive at the vehicle crash area instantly. Samadder et al. [100] proposed a sleepy motorist alert platform that utilizes an approach where the video stream application analyzes the eye blink concept by calculating the Euclidean distance of the eye and the eye aspect ratio. The IoT component generates an alert communication that includes location details, alerts through voice, and informs the vehicle owner and traffic control over the Raspberry Pi microcontroller system whenever the driver's weakness is actually recognized.

Sharaaf et al. [101] suggested an approach that concentrates on both sleepiness forecasting and crash recognition. The forecasting model is actually trained utilizing a Convolutional Neural Network (CNN), combined with a Support Vector Machine (SVM). First, the video feed of a motorist can be taken by a camera, and areas of the face, mouth, and eye can be separated. The platform can provide non-stop notifications through a smartphone program once a sleepy condition is actually recognized. The platform utilizes a two-sensor IoT-based system to detect a vehicle crash. The accelerometer and vibration modules can be utilized with the Arduino UNO microcontroller. If the motorist is unsuccessful in acknowledging the notice, an intelligent notice can be delivered to the motorist's relatives. Dange et al. [18] proposed a model that utilizes IoT and cloud connections to identify automobile crashes and inform the local EMS about the incident. This particular platform features a microcontroller, Wi-Fi component, IR module, gyroscope module, and gas sensor, among other components. The notification program actually depends on the application that informs the EMS along with the live position of the crash.

Josephinshermila et al. [102] developed a sophisticated tracking and notification platform for automobile variables, focusing on their performance and composition. Personal automobile variables are generally constantly examined through a microcontroller that saves the information on a cloud platform. The information collected includes velocity, airbag, steering, braking, and acceleration deployment data. The automobile black box process can assist with automobile safety, improve accident casualty care, support insurance companies with automobile accident inspections, and decrease fatality rates. Mohith

et al. [103] proposed a model that can recognize automobile accidents depending on monitored velocities and alert close police stations. With GPS, they can easily monitor a vehicle's speed and compare it to the velocity of other vehicles. Whenever a crash happens under suspicious conditions, an alert along with the automobile information can be released to the accountable departments.

Karthik et al. [104] develop a platform that, in the event of a vehicle accident with another automobile, detects and transmits information prepared through an Arduino microcontroller. The Arduino transmits information to a police station or a recovery team through the GSM component, which consists of location data. Once the authorities obtain the details, they can utilize the GPS component to find the area. Pradeep et al. [105] defined techniques for disaster minimization via the generation of an efficient platform, which rapidly recognizes and notifies the suitable people. The accelerometer detects potential crashes by measuring head tilt. Vibration detectors are generally capable of detecting accidents and transmitting timely alerts regarding potential dangers. Whenever the vehicle engine is turned off, the MQ3 transmits a short message to the driver's smartphone.

A crash recognition platform was created by Divi et al. [106] to identify the occurrence of a crash depending on a range of inputs, for example, a vibration module. Whenever the vehicle is actually tilted at specific angles from its initial position on the surface, the accelerometer module detects it. Then, the GSM component can transfer information showing that the automobile has been engaged in a crash, along with the GPS module to show the location. Chandra et al. [107] utilized fire and crash detection, as well as an alcohol module. If a vehicle crash happens, it transmits an immediate short message, as well as the live area, to the associated traffic controls, medical center, and fire stations. This enables the crash area to be instantly recognized and helps preserve lives.

Bhanote et al. [108] used an Arduino Uno, coupled with infrared and ultrasonic sensors, to measure the speed of oncoming vehicles closer to the accident spot. The ML algorithms, like OpenCV and CNN, were used to detect accidents in the case of a crash. As soon as the algorithms detect an accident, a message is sent to the relevant authorities on their emergency number within about five to ten seconds. Kumar et al. [109] suggested an IoT-based crash recognition and recovery platform that utilizes several detectors, such as GPS/GSM modules, MQ135 Gas module, IR detector, fire detector, DC Motor modules, and LCD screen, to identify a crash and notify the worried specialists. The suggested platform has the potential to decrease the reaction time period for a vehicle crash, thereby enhancing the possibilities of survival for the casualty.

An innovative crash recognition platform is suggested by Bhanja et al. [110] utilizing a VANET approach. Every automobile has an OBU, along with several detectors, as suggested in the research. The results of these detectors, which monitor both the motorist and the automobile, are generally input into a fuzzy logic controller for the fast recognition of a crash. Whenever a crash is actually recognized, the suggested hardware utilizes a low-range radio transceiver network to deliver an SMS to a related authority, and it can notify a nearby EMS.

Ramya Devi and Lokesh [111] concentrated on the results of an innovative crash recognition platform utilizing Vehicular Fog Computing. Utilizing the built-in detectors on a mobile phone to track vehicle accidents as well as report them to the nearest available initial emergency department, and giving live updates regarding emergencies, the chances of recovery for emergency patients would be significantly increased.

Annapoorna et al. [112] proposed a sensor system that not only identifies possible crashes but also quickly notifies rescue departments. The crash recognition system vigilantly tracks the automobile's movements, examines variations in velocity, and even uses oral hints within the automobile. Whenever a crash condition is actually recognized, the

platform triggers a notification, which can last for up to five seconds. Following this, it quickly dispatches a notice to the closest rescue group. Vijayakumar et al. [113] developed a system to detect accidents and alert personnel to provide quick medical assistance. Detectors, including the eye flicker and accelerometer module, recognize an indication of a crash and communicate the information to the connected microcontroller. The GSM module is used to determine the automobile's location (via GPS). Next, a short SMS can be delivered to the suitable emergency authority, and consequently, the EMS or relatives can quickly arrive at the site.

Joy et al. [114] introduced a versatile option regarding the improved nighttime driving safety on different roads and in remote places, while efforts were made to minimize crash intensity. The platform regularly tracks the movements of the automobile as well as its surroundings, utilizing a range of detectors, such as vibration, camera, accelerometer module, alcohol module, IR module, and a Raspberry Pi microcontroller that combine the outputs of the detectors including LCD screen, GSM, GPS, buzzer, and relay. When an accident is actually recognized, the platform rapidly logs precise areas, ranks alerts based on the severity of the accident, and sounds a buzzer to give instant notice. Kumar et al. [115] introduced an innovative automobile platform that utilizes IoT systems, incorporating multiple detectors for comprehensive vehicle tracking and monitoring. The ESP32 microcontroller processes the detector and controls the program. The platform regularly provides live automobile monitoring data and anti-theft notifications.

Vinodhini et al. [116] provided an efficient method using the LoRa technique with edge node ultrasonic detectors for discovering accidents and offering an effective transfer of EMS from the unexpected accident area to the nearest medical center with the assistance of cloud systems. Urgent details are actually transferred to the cloud, and the reaction involves notifying the surroundings and informing the suitable medical center with the assistance of a GPS sensor. Mohsin et al. [117] provided a detailed platform for implementing ML-driven emergency reaction techniques and an IoT system that utilizes live information evaluation to recognize behavior and prioritize reactions depending on their anticipated distance, emergency, and accessible information. In this regard, the XGBoost method was used to prioritize the emergencies, and its functionality was evaluated utilizing precision, average F-1 score, and recall. Furthermore, an online dashboard was implemented to visualize sensor and forecasted information live, ensuring availability and scalability. This granted people to participate with the platform via a user-friendly software and receive timely notifications and information.

Subhadra [118] developed an IoT live connection and area monitoring platform for automobile emergencies. It can transmit area details and emergency types to specific parties, for example, police departments, the nearest medical center, and family members, via SMS. The prototype of the suggested platform is designed to utilize an Arduino Uno microcontroller, GPS and GSM modules, and a MEMS accelerometer for automatic operation. Kumar et al. [119] provided a three-layer model, with the information-gathering layer depending on a low-power IoT arrangement, which consists of an MPU 6050 and a GPS module embedded into an Arduino Mega microcontroller " is manufactured by Arduino LLC (formerly Smart Projects), which is based in Ivrea, Italy.

With the substantial processing capabilities of the fog layer, the farthest cloud layer performs Multidimensional Dynamic Time Warping (MDTW) to recognize vehicle crashes, while also maintaining the databases by updating them. The experimentation compared the sophisticated algorithms, for example random forest tree, K-Nearest Neighbor, and SVM utilizing threshold-based recognition with the suggested MDTW clustering strategy.

2.4. Important Features in IoT-Based Accident Detection Study

Based on the examined research, the current literature can be generally categorized into various primary dimensions:

- Sensor Types and Deployment: Several studies, such as Kathiravan et al. [94], Mehmood et al. [95], and Oguntimilehin et al. [97], concentrate on choosing and combining sensors, including GSM, GPS, accelerometers, and ultrasonic sensors to obtain reliable accident recognition.
- Recognition Methods: Studies vary from threshold-based methods to more advanced AI techniques such as decision trees, DL classifiers, and SVMs, such as Kumar et al. [119] and Kumar et al. [7,71].
- Communication Systems: The examined research ranges from utilizing Wi-Fi and GSM modules, as seen in Rishi et al. [73], to emerging options like LoRa and Vehicle-to-Everything (V2X), as explored by Bhanja et al. [110], for sending accident notifications.
- Emergency Notification and System Integration: A range of scientific research, including Ashokkumar et al. [62], Patil et al. [64], and Sasipriya et al. [92] analyzes the utilization of IoT systems in emergency services, such as cloud-based dashboards, police departments, and EMS for live tracking.
- Environmental Considerations, Power, and Cost: Practical implementation difficulties like low cost, sensor efficiency, and battery life in different weather circumstances are regularly reviewed in some research, such as Sampoornam et al. [91], Dar et al. [48], and Kumar et al. [115].

The examined literature exhibits a dual focus on both practical applications and scientific development. Scientific research, as demonstrated by studies such as Mohsin et al. [117], Uma & Eswari [11], and Kumar et al. [119], has led to the advancement of more precise, dependable, and responsive accident recognition techniques, including multi-sensor combination, data analytics, and innovative decision-making platforms. In the case of the practical perspective, several suggested solutions, such as Vinodhini et al. [116], Subhadra [118], Pathik et al. [14], Bhanja et al. [110], and Bhanote et al. [108], are examined in simulated or real driving conditions, as well as developed to be incorporated into national safety networks, intelligent cities, or commercial automobiles. This demonstrates the growing maturity and increasing use of IoT-based platforms in improving transport safety in diverse areas and income ranges.

2.5. Accident Detection for Autonomous Vehicles

Vangala et al. [120] developed an innovative blockchain-enabled certificate-based system for recognizing automobile crashes. In this system, through the authentication procedure, every automobile is able to safely alert its nearby Cluster Head (CH) to crash-related information if a crash is recognized on highways by either its neighbor or its automobiles. Then, the CH safely transmits the information obtained from the automobiles to its roadside units, and consequently, this information is also obtained safely through the edge servers. These servers are actually responsible for planning partial blocks, including information and a digital signature on that data, as well as forwarding it to its associated server in the blockchain center for complete block development and confirmation, thereby utilizing the specific procedure.

Doecke et al. [121] predicted the safety potential of present accessible connected vehicle systems in real-world accidents. In this system, the Cohda Wireless onboard devices are generally mature connected automobile systems that were updated to achieve a low false alert rate whenever utilized in actual cases. These outcomes reveal that connected vehicle systems are significantly beneficial in real-world accident cases, and this advantage can be strengthened by having the automobile intervene autonomously with emergency

braking. Tan et al. [122] provided some proof for the accident prevention performance of systems prepared for smart and connected vehicles. In this regard, three typical techniques regarding safety benefit assessment were recognized: statistical analysis methodology, safety impact methodology, and field operation test. The disadvantages and benefits of these techniques, and also evidence for the accident prevention performance of V2V and Advanced Driver Assistance Systems (ADAS), are introduced.

Haque et al. [123] believed that there is no agreed-upon or regular technique for choosing a suitable structure for Autonomous Vehicle (AV) accident reconstruction utilizing sensor fusion. Therefore, they proposed a new simulation technique for performance monitoring. This particular research shows that a radar-camera-based centralized monitoring structure of multi-sensor usage was carried out the most among the three distinct structures examined, with different automobile accident circumstances, sampling rates, and sensor setups. Khaliq et al. [124] suggested a low-cost crash recognition and notification platform that uses an IoT-based system; primarily, it utilizes Cloud/Edge processing and V2X connection. Accordingly, automobiles are generally equipped with an OBU and mechanical sensors such as a gyroscope and accelerometer for dependable crash recognition, along with a GPS for recognition of the crash area.

Additionally, a camera is actually incorporated into the automobile to record the moment when a crash happens. Wireless systems are able to send live sensor information, but this particular function is not yet commercially accessible within the OBU of an automobile. Consequently, functional implementation is performed utilizing the IoT to generate a system among the automobiles, the central server, and the edge node. By evaluating the sufficient information obtained from highway crashes, advantageous strategies can be developed that may reduce accident fatalities.

Figure 2 illustrates a more comprehensive, structured, and hierarchical view of the IoT-based vehicle accident detection and notification system. The timeline (Figure 3) illustrates the evolution of the chosen research studies from 2008 to 2025, highlighting key review periods.

Table 1. Overview of previous studies based on the utilized sensors, findings, and limitations.

No.	Author(s)	Sensor(s)	Advantage/Limitation (L)
1	Li, et al. [20]	GPS, GSM, airbag sensors, CPS	Enhance False Alarm Rate, Time to Detect, and Detection Rate were explained right after examining the recognition operation.
2	Thompson et al. [21]	Smartphones	Decrease total traffic jams and enhance the readiness of emergency responders.
3	Chae and Yoshida [22]	GPS, GPS, LADAR, image processing	The provided prototype is actually appropriate for the avoidance of vehicle crashes.
4	White et al. [23]	Smartphones	1—A formal model for crash recognition, which is a mixed-sensor model 2—It was demonstrated how mobile phone sensors, network connections, and web services can be utilized to recognize the initial responders.
5	Zaldivar et al. [24]	Smartphones	Experimental outcomes demonstrate that the program is able to respond to a vehicle crash within 3 s, a really short time period, validating the applicability of mobile phones for enhancing safety on the highway.

Table 1. *Cont.*

No.	Author(s)	Sensor(s)	Advantage/Limitation (L)
6	Fogue et al. [25]	OBUs	Decreasing the reaction time period of EMS via the effective communication of appropriate data regarding the vehicle crash, utilizing a mixture of V2I and V2V communications.
7	Amin et al. [26]	GPS, GPRS, and GSM	Assist in enabling the rescue service to arrive in a timely manner and protect the individual's life.
8	Watthanawisuth et al. [27]	Accelerometer, GPS	Demonstrate that it is able to identify non-linear fall, linear fall, and regular ride with substantial precision.
9	Amin et al. [28]	GPS	It can save several vehicle crashers with prompt rescue.
10	Sherif et al. [29]	Speed sensor, weight sensors	The tracking station monitors the area where the vehicle crash has happened and directs injury notifications to the authorities.
11	Nazir et al. [30]	SONAR ranging modules, vibration sensor, GPS, GSM	This platform is utilized to identify the precise location of the accident and collect range data from all around the vehicle.
12	Fernandes et al. [31]	Smartphone	The mobile phone is utilized as a machine interface, allowing the motorist to install the program, receive highway risk alerts from other vehicles in the area, and terminate countdown processes upon recognizing a wrong crash.
13	Aloul et al. [32]	Smartphone	The reaction time period needed to alert emergency responders to traffic injuries can be decreased. L: Damage to the smartphone will halt all operations.
14	Sankar et al. [33]	GPS receiver, GPS receiver	Use this system to help save crucial time towards post-traumatic medical care and reduce the mortality rate.
15	Sharma et al. [34]	Smartphone	This system demonstrates that it is a less expensive option than the costly crash recognition devices installed in luxury automobiles.
16	Ibrahim et al. [35]	Motion sensor, accelerometer, gyroscope,	Quicker emergency reaction, decreased death rate and damage, and avoidance of additional crashes caused by initial crashes. L: 1—Damage to the smartphone will halt all operations. 2—It was not tested in the real environment.
17	Smolka et al. [36]	Smartphone	Rather than EMS, the family can be informed and, if necessary, they can contact the authorities. L: Damage to the smartphone will halt all operations. The results showed that the presented system will not be able to replace the eCall system completely.
18	Sany and Riyadh [37]	Vibration sensors and a buzzer	This platform provides precise location recognition of the area where the crash occurred and transmits a notification to the closest medical center and police station.

Table 1. *Cont.*

No.	Author(s)	Sensor(s)	Advantage/Limitation (L)
19	Celesti et al. [38]	Smartphone	Results show that the platform offers appropriate reaction periods, allowing motorists to receive alert messages in a timely manner, thereby preventing the likelihood of potential crashes.
20	Mohammed and Kamsani [39]	Force resistor, GPS	It was created to offer an automatic crash reporting platform as well as to save individuals' lives.
21	Zualkernan et al. [40]	Smartphone	This research presents an available, portable option that requires minimal effort from users, such as a smartphone application.
22	Khaliq et al. [42]	Inertia measurement sensor, pulse sensor, sound sensor, GPS	The program detects and analyzes the intensity of vehicle crashes with the assistance of an OBU, used in the automobile.
23	Fernandez et al. [43]	GSM, GPS	An electronic button is actually offered to cancel the transmission of a message in uncommon circumstances where there is no injury; this can save valuable time for the EMS.
24	Hadi et al. [44]	Vibration sensor, MEMS sensor	It is considered an electronic platform for instantly identifying inevitable conditions on the highway.
25	Dhanya et al. [46]	ZigBee module, MEMS sensor, vibration sensor, GPS, accelerometer, piezoelectric	If the individual in the automobile is not injured, there is actually an opportunity to cancel the notification message by pushing a button.
26	Khalil et al. [47]	Ultrasonic sensor	The suggested platform is dependable as it utilizes two ultrasonic sensors for crash recognition. L: Even though the suggested platform is entirely theoretical, the usefulness of the platform is under consideration.
27	Dar et al. [48]	Smartphone	The platform is actually the most affordable as it uses the built-in sensors in mobile phones, removing the requirement for additional equipment.
28	Vatti et al. [49]	Heart rate sensor, gyro sensor, vibration sensor, GPS/GSM	Using the reset button. The platform determines the closest police station and medical center.
29	Fanca et al. [50]	Smartphone	The platform aims to reduce false messages, guaranteeing precise recognition and dependable notifications. L: Damage to the smartphone will halt all operations.
30	Rakhonde et al. [51]	Tire pressure sensor, MQ7 sensor	The incorporation of crash recognition, crash prevention, and pollution checking in a single intelligent automobile platform utilizing IoT systems.
31	Dias et al. [52]	GPS, GSM	The platform can constantly monitor automobiles, identify crashes, and instantly alert the concerned authorities or relatives, thereby possibly saving lives by decreasing the delay in EMS.
32	Shaik et al. [41]	Accelerometer, GPS	The signal will indicate the intensity of the crash and its GPS location. The EMS can utilize the GPS module to arrive in the field rapidly.

Table 1. *Cont.*

No.	Author(s)	Sensor(s)	Advantage/Limitation (L)
33	Nanda et al. [53]	GPS, GSM, vibration sensors, accelerometers	Delivering an SMS to a nearby medical center and considering whether the motorist is actually sleepy or in an unpredictable condition.
34	Mankar and Tasgoankar [54]	GPS, GSM, accelerometer	Instantly provides the crash site location to the rescue group so that they can offer medical assistance to the casualty as early as possible to save lives.
35	Al Wadhahi et al. [45]	IR sensors, GSM, GPS	Alert the motorist regarding the nearby automobiles whenever the distance between them exceeds the threshold magnitude.
36	Taj et al. [55]	Vibration sensor, GSM, GPS	The platform provides an SMS to a close rescue group and police station through the GSM component.
37	Bhatti et al. [6]	Smartphones	A personalized Android program is established to collect data concerning velocity, pressure, gravitational force, audio, and area. L: Damage to a smartphone will halt all operations.
38	Wang et al. [68]	Smartphone	Prevents rear-end accidents whenever following up at low velocity, backward accidents whenever reversing, as well as side accidents due to the blind spot.
39	Kashevnik et al. [69]	Smartphone	This system facilitates the following motorist's hazardous conditions: sleepiness, distraction, and an offline hazardous condition associated with a high pulse rate.
40	Kumar et al. [70]	Smartphone	The IoT technique aims to offer an available and most affordable option for precise crash recognition and quick emergency reaction utilizing standard mobile phone detectors, improving automobile safety, and possibly saving lives.
41	Dar et al. [56]	Smartphone	Lower reaction times and reduced cost. L: Damage to the smartphone will halt all operations.
42	Devi and Pamila [57]	Smartphone	This platform can effectively centralize information, increase automation, enhance protection, as well as support seamless integration, thereby enhancing functionality and decreasing expenses. L: Damage to the smartphone will halt all operations.
43	Sarker et al. [58]	Accelerometer sensor, ultrasonic sensor, GPS, GSM	The system is actually affordable and straightforward to set up inside any vehicle.
44	Hassan et al. [59]	Vibration sensor, accelerometer,	The main advantages of this particular platform are that it is affordable, safe, and simple to utilize. L: Damage to the smartphone will halt all operations.
45	Shankarpure and Abin [60]	Smartphone	The platform can share the precise area of the crash with urgent medical services. L: Damage to the smartphone will halt all operations.

Table 1. *Cont.*

No.	Author(s)	Sensor(s)	Advantage/Limitation (L)
46	Chang et al. [61]	MEMS sensor, GPS module, front camera	The precision of traffic accident recognition can achieve 96%, as the average reply period for emergency reaction is around 7 s.
47	Ashokkumar et al. [62]	vibration sensor, alcohol detection, eye blink sensor, web cam	If the individual has a minor crash or if there is no severe danger to any person's life, the motorist can cancel the notification message through an electronic button to prevent wasting the EMS's precious time.
48	Choudhury et al. [63]	Accelerometer, heart rate sensor, GPS, GSM	This particular platform can ensure the fastest arrival of medical support, providing casualty with a better chance of survival.
49	Patil et al. [64]	Smartphones	The system rapidly finds vehicle crashes as well as notifies EMS, possibly saving lives by decreasing reaction times using GPS and IoT integration. L: Damage to the smartphone will halt all operations.
50	Gowri et al. [65]	GPS, GSM, vibration sensor, heart rate sensor, eye flicker sensor	This research aims to prevent vehicle crashes caused by the fatigue of the motorist and to inform the motorist's family members in the event of a crash or abnormal circumstances.
51	Kader et al. [66]	Speed sensor, accelerometer, GSM	The authorities can track driving quality details from anywhere in the world via the internet. Consequently, this platform can considerably increase the liability of the motorist to prevent careless driving.
52	Habib et al. [78]	Accelerometer, vibrating sensors	The purpose is actually to assess the extent of the crash site and determine the severity of the injury, so that a suitable level of assistance can be offered as quickly and effectively as possible, while minimizing traffic congestion on the highway.
53	Kashevnik et al. [79]	Smartphone	A mobile phone-based platform offers the advantages of mobility and low cost compared to built-in devices. As mobile phones are generally personal, it is simple to adapt and train. L: Damage to the smartphone will halt all operations.
54	Kumar et al. [71]	Smartphone	The platform can be utilized in any automobile to decrease the period of automatic notice right after the accident. L: The platform needs a constant internet connection to transmit emergency notifications. L: The placement of the mobile phone would be predetermined, as users are unable to put the mobile phone in any other place, such as a bag or a pocket. L: If the mobile phone gets ejected outside the automobile or hardware installation breakdowns occur, outcomes can be influenced, and the platform can fail.
55	Karmokar et al. [72]	Load cell, GPS, GSM	The proposed system can provide the predicted result in a relatively cost-effective way.

Table 1. *Cont.*

No.	Author(s)	Sensor(s)	Advantage/Limitation (L)
56	Rishi et al. [73]	GPS, GSM, accelerometer	L: This platform cannot transmit data to the relatives, medical center, and government companies to generate information on crashes.
57	Ajao et al. [74]	GPS, GSM, accelerometer, gyroscope sensor, radar sensor, carbon monoxide sensor, IR sensor	Alerts the motorist prior to an accident occurring and eliminates the delay period between the crash and the arrival of emergency personnel.
58	Rana et al. [75]	GPS, GSM, accelerometer, ultrasonic sensors	Simple navigation to the crash site.
59	Kapilan et al. [76]	GPS, GSM, shock sensor, smoke sensor, accelerometer, gyroscope	It can assist the medical backup team in arriving at the crash location in a timely manner, potentially saving valuable lives. L: This platform cannot recognize the location of the automobile or the specific part of the automobile that was damaged.
60	Aung and Thein [77]	Smartphones	Even if the driver is actually unconscious, the platform can wait for a specific time period before contacting the closest medical center. L: Damage to the smartphone will halt all operations.
61	Sampoornam et al. [91]	GPS, GSM, accelerometer, drowsiness detector, tilt sensor	The main strengths of this particular platform are its affordability, guaranteed safety, low power usage, the ability to save a casualty's life rapidly, increased precision, and reduced possibility of individual error.
62	Balfaqih et al. [13]	Heart rate sensor, GPS, vibration sensor, accelerometer, flame sensor, smoke sensors, force sensor, impact sensors	This platform demonstrated that the Classification and Regression Trees (CART) and Gaussian Mixture Model (GMM) models performed significantly better in terms of accuracy and recall.
63	Sasipriya et al. [92]	Smartphone	This system can help the EMS area stay updated in the repository. Furthermore, the traffic signal can be managed in accordance with the future EMS, consequently offering a traffic-free route. L: Damage to a smartphone will halt all operations.
64	Priyath [93]	Smart watches, grip pressure, heart rate, speed sensor	The multi-sensory information, as well as the fuzzy model, provide valuable data for sleepiness recognition.
65	Bhakat et al. [12]	Accelerometer, gyroscope, camera	L: Considers only the nearest hospitals and police stations, disregarding traffic congestion levels.
66	Rehman, et al. [80]	Tilt sensor, GSM, gyroscope, accelerometer, force sensor	The created platform has been broadly examined in real-time circumstances. L: The system cannot determine the severity of the accident.
67	Chikaka & Long [81]	Accelerometer	A reset switch has been included to deactivate the platform in the event of a minor crash.
68	Yellamma et al. [82]	GPS, GSM, ultrasonic sensor, accelerometer sensor	L: Sends a limited message to the family.

Table 1. *Cont.*

No.	Author(s)	Sensor(s)	Advantage/Limitation (L)
69	Babalola & Olokun [83]	GSM, GPS, vibration sensor, MEMS sensor	This platform can be utilized for both rollover accidents and accident severity.
70	Narayanan et al. [84]	GPS, GSM, vibration sensor	Assists in saving the individuals who were involved in a crash and notifies the police station and insurance company.
71	Chaithanya et al. [85]	GPS, GSM, accelerometer, eye blink sensor, pulse sensor	When a crash is actually recognized, the buzzer sounds. If the motorist does not react to the buzzer within a fixed period, the system will activate. L: Damage to the smartphone will halt all operations.
72	Alkinani et al. [86]	Smartphone	The suggested platform demonstrates promising outcomes in terms of precision and reaction time compared to current methods. L: Damage to the smartphone will halt all operations.
73	Kumar et al. [87]	Accelerometer, GPS, temperature sensor, heart rate sensor, MEMS sensor	Allowing quicker reactions from EMS to the crash area and tracking people who try to run away from the location of a crash. L: Only a simulation model.
74	Sumathy et al. [88]	GPS, GMS	If the victim is not severely injured, they can turn off the notification process by pressing a switch located on the platform.
75	Mahadik et al. [89]	Smartphones	Provides on-time details to the EMS concerning the area of the crash, which helps to save an important life. L: Damage to the smartphone will halt all operations.
76	Kathiravan et al. [94]	Acceleration sensor, GPS, Vibration sensor, gas Sensor	The platform can help people in various ways, such as offering rapid support in the event of a crash, monitoring the vehicle in the event of theft, and remotely deactivating the automobile.
77	Mehmood et al. [95]	GPS, GSM, ultrasonic sensor, camera	The platform described the severity of the crash, whether an automobile had been involved in an accident with another automobile or a disaster had occurred to the automobile itself.
78	Tippannavar et al. [96]	GPS, MQ-3 sensor, bump sensors, accelerometer	To alert EMS of the crash and to monitor and identify the precise area of any vehicle that has been involved in an accident.
79	Oguntimilehin et al. [97]	Vibration sensor, tilt sensor, flame sensor, and GPS	The platform can send the details with a 45 s delay, allowing the motorist to reset the program if a crash is mistakenly identified. Otherwise, the details regarding the crash can be delivered, and the medical center nearest to the crash site can be identified.
80	Selvi et al. [98]	Vibration sensor, MEMS sensor, accelerometer sensor	The platform is helpful in providing a fast response to crashes and may substantially decrease fatality rates.

Table 1. *Cont.*

No.	Author(s)	Sensor(s)	Advantage/Limitation (L)
81	Tamilselvan et al. [99]	Gyro sensor, fire sensor, alcohol and sound sensor, sensor data from the automobile	This system can be beneficial in crash-prone areas at an affordable price, while also generating revenue for government organizations to initiate such systems and additional services.
82	Samadder et al. [100]	Camera	The recommended platform excels in that it is able to recognize tiredness during nighttime and daytime, with roadblocks at various ranges, with a precision higher than 98%.
83	Dange et al. [18]	Gas detection sensor, gyroscope sensor, and IR sensor	The notifying platform depends on an application that informs the EMS of the live area of the crash.
84	Josephinshermila et al. [102]	Temperature sensor, gas sensor, DC motor and IR sensor, GPS, GSM	This platform can help improve automobile safety, enhance care for accident victims, assist insurance companies with automobile accident inspections, and enhance highway conditions to decrease fatality rates.
85	Mohith et al. [103]	GSM, GPS, vibration, IR sensor, ultrasonic sensor, gas sensor	There is no mistake-tracking system in place to avoid accidents from occurring on this platform.
86	Karthik et al. [104]	GSM, GPS, Accelerometer, vibration	The system notifies the EMS and the relatives for immediate assistance by providing the location whenever a crash occurs.
87	Divi et al. [106]	Vibration sensor, accelerometer sensor, GSM, GPS, camera	If a picture of the surroundings is delivered, the crash location can be quickly identified. The camera component transmits a picture of the crash site surroundings.
88	Chandra et al. [107]	GSM, GPS, MQ3 sensor, flame sensor, ultrasonic sensors	This system can be used for college and school vehicles to ensure the safety of students and staff. It locates fire and sprinkles water, decreasing fire crashes and protecting essential life.
89	Bhanote et al. [108]	Ultrasonic sensors	Once the system identifies a crash, an alert can be delivered to the appropriate authorities on their pre-determined number.
90	Kumar et al. [109]	GSM, GPS, fire sensor, IR sensor, gas sensor	The suggested platform has the potential to decrease the reaction period in the event of a crash, thereby enhancing the victim's chances of survival.
91	Bhanja et al. [110]	Low-range radio (LoRa) transceiver network	After obtaining the information from an automobile, the system can notify a nearby EMS. It can determine the quickest route to a nearby medical center utilizing Dijkstra's algorithm and notify the EMS.
92	Ramya Devi and Lokesh [111]	Smartphone	This system ensures the optimization of traffic flow and energy usage during crashes, as well as in foggy conditions.
93	Annapoorna et al. [112]	Vibration sensor, accelerometer, ultrasonic sensors, and IR sensor	The suggested system indicates a superior and complete strategy for addressing crashes.

Table 1. *Cont.*

No.	Author(s)	Sensor(s)	Advantage/Limitation (L)
94	Vijayakumar et al. [113]	Accelerometer, eye flicker sensor, GPS, GMS	A message can be sent to the appropriate emergency responders, allowing the rescue vehicle or relatives to quickly arrive at the scene.
95	Joy et al. [114]	Accelerometer, camera, vibration sensor, IR sensor, alcohol sensor, GPS, GSM	The platform, which easily integrates into current automobile devices, aims to reduce crash intensity and enhance results with a focus on emergency reaction periods.
96	Kumar et al. [115]	Alcohol detection, GSM, GPS, eyeblink sensor, vibration sensor	It was developed to be innovative, most affordable, and simple, making it a helpful asset for any motorist.
97	Vinodhini et al. [116]	Ultrasonic sensors, GPS	Urgent details are actually transferred to the cloud, and the reaction involves notifying the surroundings and informing the suitable medical center with the assistance of a GPS sensor.
98	Mohsin et al. [117]	GIS, Cloud computing	This particular platform not only enhances instant reaction abilities but also helps in proper planning through offering valuable insights into possible upcoming events.
99	Subhadra [118]	GSM and GPS modules, MEMS accelerometer	It can send area details and emergency forms to the appropriate recipients, for example, relatives and family, the nearest medical center, and police departments via a text message.
100	Kumar et al. [119]	GPS, MPU 6050, placed on an Arduino Mega	The analysis compared innovative algorithms, for example, the random forest tree, K-Nearest Neighbor, and SVM.
101	Vangala et al. [120]	Blockchain technology	Due to the utilization of blockchain systems, it is demonstrated that this system is not only safe from various potential attacks but also maintains decentralization, immutability, and data transparency.
102	Doecke et al. [121]	Cohda wireless onboard units, GPS	The outcomes reveal that connected vehicle systems can be significantly helpful in real-world accident situations.
103	Tan et al. [122]	Connected vehicle	Technologies on ICVs could considerably decrease the number of accidents.
104	Haque et al. [123]	Radar-camera-based, multi-sensor fusion	The SMTPE assists in choosing the most extraordinary monitoring architecture for AV accident reconstruction.
105	Khaliq et al. [124]	On-Board unit, accelerometer, gyroscope, GPS, camera module	By conducting an evaluation of the sufficient information obtained from highway crashes, advantageous strategies for action which may limit accident fatalities can be developed.

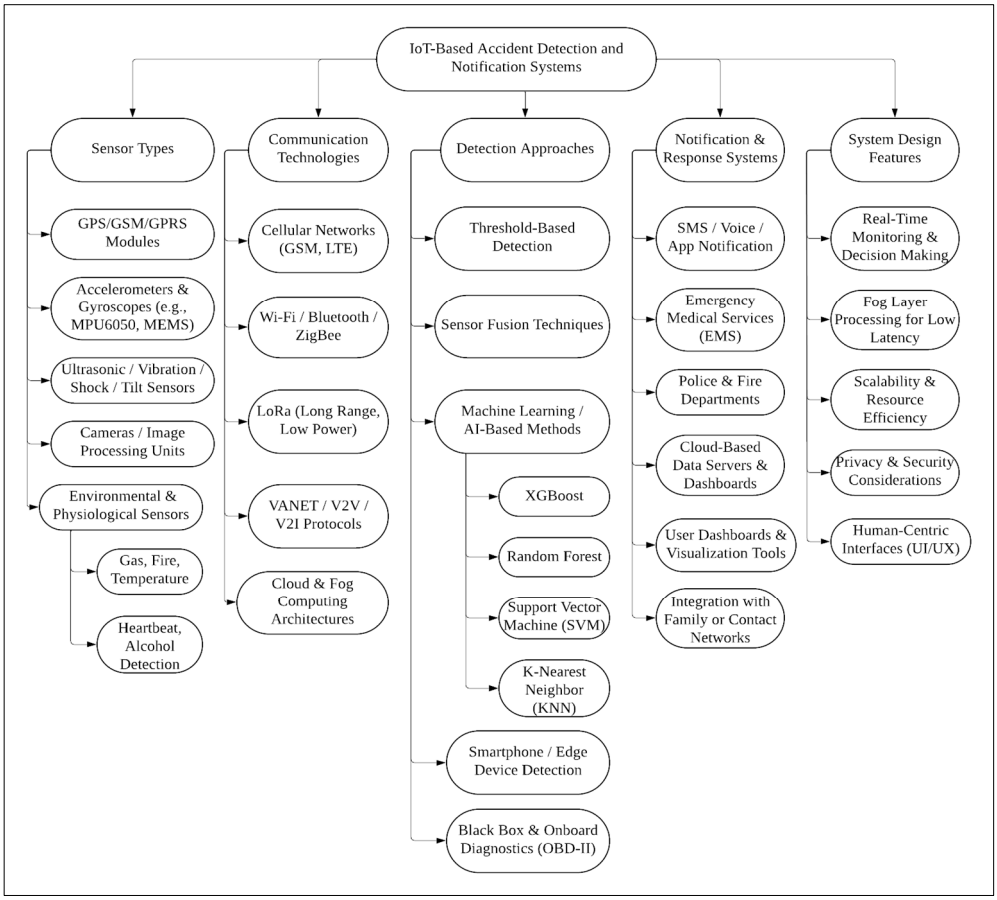

Figure 2. Classification of IoT-based vehicle accident detection and notification systems.

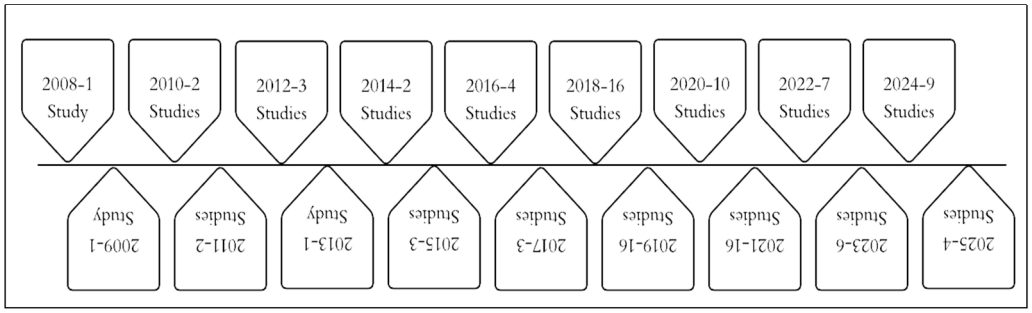

Figure 3. The Timeline of the IoT approaches for accident detection and emergency notification from 2008 to 2025.

3. Research Methodology

This particular research utilizes a comparative assessment strategy to evaluate IoT-based crash recognition and notification systems. The aim was to identify, categorize, and examine the various types of sensors utilized in these platforms, as well as to recognize crucial study trends over time.

All studies were chosen depending on specific key phrase searches in well-known scientific databases, such as ScienceDirect, SpringerLink, IEEE Xplore, and Scopus. Key phrases provided combinations of emergency reaction, smart sensors, accident notification, vehicle accident detection, and IoT. Studies were included if they (1) used recognizable modules or sensors, (2) considered IoT-based crash recognition or emergency alert systems, (3) were released between 2008 and 2025, and (4) presented specialized information on

system architecture or performance. A total of 101 scientific research studies were listed after filtering in the ultimate evaluation.

To conduct a comparative evaluation, the sensor types utilized in each research were manually identified and saved in Excel software. The information involved almost 40 unique sensor types, for example, accelerometers, GSM, GPS, gyroscopes, smartphones, vibration sensors, cameras, IR sensors, and ultrasonic sensors. A radar graph, as shown in Figure 4, was created from this particular data to visually demonstrate the variety and frequency of sensor utilization throughout the entire research.

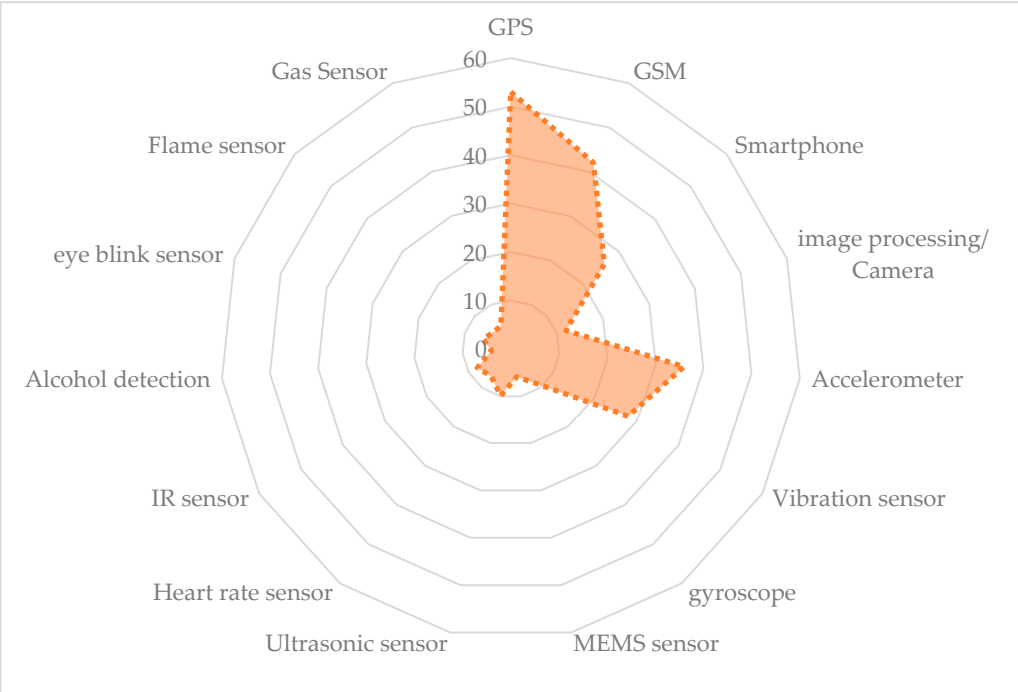

Figure 4. The overall number of different sensors in the prior study.

Figure 4 obviously displays that accelerometers are utilized in 36 research studies, GSM in 42 research studies, while GPS is used in 53 research studies, and they are generally the most frequently applied elements in IoT-based accident recognition systems. Other frequently used sensors include ultrasonic sensors, smartphones, gyroscopes, vibration sensors, and image processing/cameras. These outcomes highlight the industry's concentration on multi-sensor utilization as well as live geolocation monitoring.

Although a year-wise evaluation is briefly provided, a much more detailed cross-country evaluation associated with motorization degrees is certainly recommended for upcoming research. This type of evaluation will undoubtedly improve the comprehension of territorial adoption trends, as well as the scalability of IoT solutions for various transport infrastructures.

4. Comparative Analysis and Discussion

4.1. Commonly Used Sensors in IoT-Based Accident Detection Systems

The combination of GSM (42 studies) and GPS (53 studies) modules in vehicle crash recognition platforms shows their essential function in live location monitoring as well as emergency connection. The GPS module allows accurate vehicle crash localization, guaranteeing that emergency authorities are able to rapidly arrive at the accident area, while GSM helps immediate notifications through smartphone networks. Nevertheless, these types of systems are significantly reliant on network accessibility, making them ineffective in distant or blocked locations, for example, compressed urban environments

or tunnels. To improve vehicle crash recognition precision, vibration modules (28 studies) and accelerometers (36 studies) were extensively utilized. These types of sensors identify unexpected movement impacts or changes, delivering an instant signal of an accident. In spite of their performance, they are generally susceptible to false alarms due to potholes, sudden braking, or highway problems, necessitating the incorporation of extra detectors for greater dependability.

Other frequently utilized sensors consist are ultrasonic (ten studies) and infrared (eight studies) sensors, which enhance vehicle crash avoidance through identifying obstructions and automobile distance. Although infrared modules are invaluable in low-visibility circumstances, their range restrictions need extra help from ultrasonic modules that offer accurate range measurement in accident or parking situations. Furthermore, tilt sensors (three studies) and gyroscopes (nine studies) perform an essential function in discovering rollovers, particularly for motorbikes and trucks. In motorist tracking platforms, heartbeat sensors (seven studies) as well as alcohol recognition modules (four studies) play a role in avoidance and post-crash evaluation, guaranteeing that medical aid is actually offered whenever required. Nevertheless, these kinds of physical detectors need direct user interaction that can restrict their performance in live situations. As crash recognition investigation progresses, the collaboration of multi-sensor incorporation with AI algorithms is becoming an increasingly important trend to reduce false alarms and enhance vehicle crash reaction efficiency.

Figure 4 shows the total quantity of various sensors used in previous studies, particularly concentrating on those that were used more than three times. Other sensors, which were used less frequently, include the airbag sensor, CPS module, LADAR, OBUs, speed sensor, weight sensor, SONAR ranging, inertia measurement, sound sensor, piezoelectric sensor, tire pressure sensor, MQ7 sensor, radar sensor, shock sensor, grip pressure, and bump sensor, which have been utilized in one study each. In addition, a puls sensor, a ZigBee/Xbee module, a speed sensor, a smoke sensor, and a temperature sensor were used in two studies, while a force resistor and a tilt sensor were utilized in three studies.

The preference for specific types of sensors, for example, GSM and GPS, is strongly associated with their cost efficiency, maturity, and simplicity of use in automobile products. These types of elements are generally recognized across different hardware systems and provide steady functionality even in low-resource environments. The excessive utilization of the accelerometer sensor displays its dependability in accident recognition circumstances where motion sensitivity is generally significant. On the other hand, more recent sensors, such as biometric sensors, ultrasonic modules, and gyroscopes, are increasing attention; however, they remain underutilized due to their complicated usage, calibration needs, and restricted service in lower-end products. This particular gap presents a possible area for upcoming advancement, particularly in improving sensor fusion platforms for enhanced precision and contextual awareness.

4.2. Comparative Analysis Based on the Years

The implementation and progress of IoT systems in vehicle crash recognition, as well as alert platforms, have substantially developed in recent times, with several experts contributing effective methods. This evaluation provides a comprehensive analysis of research based on the year of publication, highlighting its benefits, strengths, and limitations.

4.2.1. Early Contributions (2008–2010)

From 2008 to 2010, early IoT-based vehicle crash recognition platforms concentrated on establishing basic sensors for real-time accident identification. Li et al. [20] developed a system that combines GSM and GPS sensors for live area monitoring, although it relies on

different hardware, such as airbags. Thompson et al. [21] developed mobile phone-based recognition platforms that recognize false positives via smartphone context data, providing a significant enhancement in functionality and dependability. Chae and Yoshida [22] investigated RFID systems for vehicle crash avoidance in heavy machinery, establishing a prototype to improve live risk recognition. In general, these kinds of research set the groundwork for upcoming developments; however, they confronted problems, such as hardware restrictions, false alarms, and the requirement for consistent protocols.

4.2.2. Mid-Term Advances (2011–2015)

White et al. [23] and Zaldivar et al. [24] leveraged mobile phone systems to present functions, for example, connection and accelerometer for vehicle crash recognition. These types of research have considerably contributed to decreasing reaction times in emergencies; however, they have encountered restrictions because of the sensitivity of mobile phone sensors. Likewise, Fogue et al. [25] incorporated V2I and V2V communications to improve vehicle crash recognition precision and emergency reaction performance, although costly onboard devices was required.

Improvements were achieved between 2013 and 2015 by utilizing multimodal alarm distribution (Fernandes et al. [31]) and ML algorithms regarding intensity evaluation (Aloul et al. [32]). These techniques marked a shift toward more advanced forecasting platforms, addressing concerns about false positives and scalability. Nevertheless, computational needs, as well as implementation expenses, continued to be issues.

4.2.3. Recent Innovations (2016–2023)

Sankar et al. [33] suggested the use of extensive vehicle crash and EMS management platforms utilizing Android applications, presenting capabilities for quick dispatch. Dar et al. [48] created a fog computing system for live emergency reactions, decreasing latency in information transmission. Likewise, Choudhury et al. [63] used Arduino programs to incorporate heartbeat tracking with vehicle crash recognition, showing a shift toward comprehensive tracking.

Research within 2020 significantly integrated DL and AI, as seen in Kumar et al. (2020) as well as Pathik et al. [14] who used cognitive algorithms for improved precision in vehicle crash classification and recognition. The incorporation of IoT systems with cloud computing, as utilized by Dange et al. [18], enables efficient information sharing as well as emergency notices.

4.2.4. Emerging Trends and Future Directions

Latest studies, for example, those by Bhanote et al. [108] and Kumar et al. [109], have highlighted the utilization of multiple sensors, such as IR sensors, gas sensors, and ultrasonic sensors, to enhance recognition reliability. The utilization of ML and fuzzy logic methods (e.g., Bhanja et al. [110]) signals an upcoming trend toward smart and autonomous automobile platforms. In spite of these kinds of developments, issues, for example, scalability, expense, and interoperability across platforms continue, needing additional research.

The development of an IoT-based vehicle crash recognition platform shows a progressive shift from simple sensor incorporation to advanced, as well as AI-driven options. However, earlier studies concentrated on basic systems; the latest research has conducted multi-sensor systems. Upcoming study needs to highlight interoperability, cost-effectiveness, as well as worldwide standardization to fully understand the possibilities of these types of platforms.

Sensor usage over the years has shown a substantial development from basic, threshold-based recognition platforms to more enhanced, AI-powered designs. Initial

research mainly depended on GSM and accelerometer sensors for simple accident recognition and notification. Nevertheless, the growth of mobile phones, the use of ML methods, and the expansion of cloud computing since 2015 have led to an increase in the usage of innovative problem-solving platforms. These developments are particularly notable in nations with strong digital facilities, highlighting a clear link between research development and technological readiness. Current research also focuses on live communication, multi-source information utilization, and automated notification platforms, all of which are arranged with the response needs of emergency reaction providers and promote more intelligent mobility.

4.3. Comparative Analysis Based on the Used Sensors

4.3.1. Vision-Based and Video Processing Systems

Vision-based vehicle crash recognition relies on image processing to identify and monitor traffic accidents. Ki and Lee [125] established a platform that utilizes digital cameras to extract automobile motion and identify anomalies depending on directional and speed changes. Likewise, Choi et al. [126] used dashboard digital cameras and DL models to enhance accident recognition precision. This method offers several advantages, including the ability to successfully investigate complex vehicle crash circumstances, and does not require an in-vehicle setup. It has some disadvantages, including restricted performance in low-visibility circumstances such as undesirable weather and nighttime driving.

4.3.2. Smartphone-Based Detection Systems

With the growth of mobile phones, several researchers have utilized built-in detectors for vehicle crash recognition. White et al. [23] and Aloul et al. [32] used GPS and accelerometer information to identify abrupt effects as well as alert emergency authorities. Mahadik et al. [89] suggested working with mobile phone programs. This provides timely details of emergency providers concerning the area of the vehicle crash, which helps to protect people's lives. Although these systems utilize several sensors, such as accelerometers, gyroscopes, GPS, and microphones, they have some advantages and limitations. The advantages include low cost, high availability, and the fact that no extra hardware is needed. On the other hand, it has some issues with the sensitivity of smartphone sensors and requires user input for optimal accuracy.

4.3.3. IoT and Sensor-Based Systems

IoT-based platforms incorporate several detectors with connection systems for live crash recognition and notification. Fogue et al. [25] created a platform that utilizes V2I and V2V connections to improve recognition precision. Lately, Pathik et al. [14] established an AI-powered IoT platform able to discover vehicle crash intensity depending on multiple weather conditions. The most notable advantages include high precision, quick response times in emergencies, and the ability to combine multiple sensors, all of which contribute to more reliable detection. Disadvantages include the need for special hardware, high cost, and reliance on network connectivity.

4.3.4. Driver-Monitoring Systems

The driver-monitoring platform's goal is to identify potential vehicle crashes through monitoring motorist actions and physical indicators. Sharaaf [101] and Priyath [93] established a wearable recognition platform that tracks driver behavior and sleepiness via heartbeat and eye-tracking detectors. The advantages of these systems include early crash avoidance, improved motorist safety, as well as incorporation with vehicle-based platforms, while the limitations are that they need user compliance and can generate false alerts.

4.3.5. AI and Machine Learning-Driven Systems

The latest research has concentrated on AI-driven models to enhance vehicle crash recognition precision. Bhanja et al. [110] suggested a fuzzy logic method for VANET, though Chang et al. [61] used DL to sort out crash intensity. The advantages of this approach are high accuracy, ability to predict the severity of an accident, and capacity to adapt in real time. The disadvantages are that it needs a lot of computing power and relies on big training datasets.

Besides determining the varieties of sensors utilized in distinct research, it is crucial to take into account the broader technological framework in which these kinds of platforms have developed. For example, nations with superior manufacturing abilities and digital facilities, like several European Union countries, the United States, South Korea, and Japan, are generally more likely to apply complicated sensor fusion methods, such as AI-powered and GPS-integrated accelerometers. The accessibility of 4G/5G systems, as well as cloud computing systems, in these areas has facilitated the processing of live information, remote emergency notifications, and integration with national transportation data sources. On the other hand, research from growing economies often concentrates on less complicated recognition techniques that utilize the most affordable sensors, such as basic accelerometers and GSM, because of financial and funding constraints. These distinctions reveal how investment in digital infrastructure, motorization rates, and national advancement ranges shape the deployment and advancement of IoT-based crash recognition systems. Recognizing these kinds of disparities is crucial for analyzing the scalability and feasibility of suggested platforms across distinct areas. In general, the distribution of sensor combinations in several research studies signifies different ranges of technological capability and functionality concentration. For example, combinations including accelerometer, GSM, and GPS in areas with trustworthy network coverage, as well as reasonable costs, provide a balance of affordability and accuracy. Considerably more complicated setups, including environmental sensors, gyroscope sensors, or cameras, are generally observed in several research studies from institutional research labs or technologically advanced nations. These kinds of designs help superior accident identification; however, they cause difficulties regarding information processing requirements, power usage, and cost. The variations in sensor usage, furthermore, reveal regional priorities, which include areas concentrating on safety in high-traffic city regions, though others prioritize low-cost options for rural areas.

4.4. Critical Synthesis and Research Gaps

A comparison synthesis of the reviewed systems shows that GSM and GPS components remain the most extensively utilized elements, mainly because of their low cost, dependability, and ease of integration into current automobile platforms. Vibration and accelerometer sensors are generally prevalent, providing fast accident recognition abilities depending on motion anomalies. New research shows the need to expand attention on integrating sensor fusion and AI-driven platforms to enhance precision; however, such platforms frequently incur higher costs and increase deployment complexity.

There is a significant trade-off among implementation feasibility, precision, and cost, especially in research focusing on deployment in rural or low-income areas. Although some plans accomplish high recognition accuracy, they frequently depend on the need for a robust network infrastructure or costly multi-sensor arrays that may not be accessible in underdeveloped nations.

Significant research gaps consist of limited real-world or large-scale examination, the lack of standardized functionality metrics, and inadequate concern for data privacy and safety measures, particularly on platforms utilizing biometric and GPS information.

Furthermore, most affordable and scalable alternatives remain underexplored, in spite of their significant influence in developing countries.

From a historical viewpoint, the field has developed from early GSM-based and RFID platforms in the 2000s toward multi-sensor and AI-powered systems supported by cloud and edge computing. This particular advancement displays not only technical development but also a switch in research focus from basic recognition to strong, live emergency reaction.

4.5. Simulation-Based Approaches in Accident Detection and Prevention

Recently, simulation tools, for example, SUMO, OMNeT++, and CARLA, as well as integrated co-simulation environments, have performed a significantly crucial function in establishing and validating crash recognition and avoidance systems.

In this regard, Tolba and Kamal [127] showed an improved model for the SDC-Net program by (1) adapting the benchmark dataset labels developed on the CARLA simulator to involve the automobiles' bounding boxes though maintaining the same training, testing, and validation process, (2) changing the classification system with a recognition platform, and (3) changing the shared data through IoT to involve the crash area. The SDC-Net++ program is actually suggested to (1) output the appropriate control actions, particularly during crashes, and (2) share the most essential data to the connected automobiles through IoT systems, particularly the crash site location. The SDC-Net++ multitask system with bird's eye view (BEV) outperforms SDC-Net multitask with BEV in accuracy. Bajpai et al. [128] demonstrated a functional simulation design utilizing SUMO and OMNET++ to create inter-vehicle connections for suitable traffic management, as well as road safety regarding road traffic crashes. The generation of network and traffic simulators, such as SUMO and OMNET++, is generally linked to the "VEINS" system, an open-source platform for simulating the operating systems of automobiles. The result is actually intended to enable several automobiles to connect generally, as well as on a manually created map through information packet transfer on particularly triggered events.

Beg and Ismail [129] examined automobile recognition at roundabouts, cross-junctions, and priority junctions utilizing image information received through the CARLA system. Subsequent evaluation distinguishes between non-vehicle objects and vehicles in the dataset. They assessed the likelihood of vehicle crashes by monitoring important elements such as automobile velocity, density, and distance on different road types. Advanced ML methods are suggested to examine the performance of the automobile recognition platform in gathering important variables such as distance, velocity, and vehicle count in roundabout and junction scenarios. Chen et al. [130] suggested a new virtual–real-fusion simulation platform, which integrates vehicle crash creation, unmanned aerial vehicle-based image selection, as well as a 3D traffic crash reconstruction pipeline with unsupervised 3D point cloud clustering systems and enhanced computer vision methods. Particularly, an autonomous driving simulator and a micro-traffic simulator are generally co-simulated to create highly precise vehicle crashes. Eventually, a DL-based reconstruction technique, specifically 3D Gaussian splatting (3D-GS), is actually used to develop 3D digitized vehicle crash moments from UAV-based photo datasets gathered in the traffic simulation environment. A mixed-integer programming Bayesian optimization algorithm and a clustering parameter stochastic optimization method are suggested to improve the segmentation of large-sized 3D point clouds. In statistical tests, 3D-GS generates excellent, seamless, and live-rendered vehicle crash scenes, obtaining a structural similarity index of up to 0.90 within various cities.

4.6. Public Datasets for Accident Prediction and Model Development

Caesar et al. [131] introduced the initial dataset, known as nuTonomy scenes (nuScenes), which comprises a comprehensive autonomous automobile sensor collection—one lidar, five radars, and six cameras, almost all with a total field of view of nearly 360 degrees. This dataset includes 1000 scenes, each 20 frames long, as well as is completely annotated with 3D bounding boxes regarding eight attributes and twenty-three classes. It also includes seven times as many annotations, as well as 100 times as many photos as the pioneering KITTI dataset. They offer careful dataset evaluation and baselines regarding image and lidar-based recognition and monitoring. Sun et al. [132] created a novel, diverse, high-quality, and large-scale dataset. Their new platform includes 1150 moments that span 20 s, featuring properly coordinated and calibrated professional camera and LiDAR information taken over a variety of suburban and urban areas. It is actually 15 times more varied compared to the biggest camera + LiDAR dataset accessible in accordance with their suggested diversity metric. Sun et al. [132] extensively annotated this information with 3D (LiDAR) and 2D (camera image) bounding boxes, with constant verifications over frames. This database offers robust baselines for 3D and 2D recognition and monitoring projects.

Chang et al. [133] offered a dataset developed to assist autonomous automobiles, known as Argoverse, consisting of motion prediction and 3D monitoring. This dataset consists of sensor information gathered through a fleet of autonomous automobiles in Miami and Pittsburgh, 3D monitoring annotations, 300,000 extracted interesting automobile trajectories, as well as rich semantic maps. In this regard, the sensor information includes 3D point clouds from a long-range LiDAR and 360-degree photos captured through seven cameras. Their 290 km of mapped lanes consist of semantic metadata as well as rich geometric data that are not currently accessible in any open dataset. Motion prediction experiments ranging in complexity from traditional techniques such as k-NN to LSTMs illustrate that utilizing specific vector maps with lane-level data considerably decreases forecasting error. Wang et al. [134] offered a large-scale dataset, known as DeepAccident, created through a realistic simulator that encompasses various crash situations commonly encountered in real-world traveling. This dataset consists of 285,000 annotated samples and 57,000 annotated frames, approximately seven times greater than the large-scale nuScenes dataset, which contains 40,000 annotated samples. Furthermore, Wang et al. [134] suggested a novel process, end-to-end motion, as well as crash forecasting that can be utilized to directly examine the crash forecasting capability of distinct autonomous traveling algorithms. Moreover, for each situation, they set up four automobiles along with one infrastructure to capture information, thereby offering different viewpoints for crash situations, as well as allowing V2X investigation on forecasting and perception tasks.

5. Policy Suggestion and Limitation

5.1. Policy Suggestions

To improve the performance of IoT-based vehicle crash recognition platforms, policymakers need to consider both regulatory systems as well as standardization regarding sensor incorporation within automobiles. Developing instructions for setting up essential detectors, for example, processing cameras, accelerometers, and GPS, in private and commercial automobiles can considerably enhance highway safety. Authorities must present policies requiring producers to install intelligent sensors in recently developed automobiles, whilst offering funds for older automobile owners to enhance their platforms. Furthermore, international and national transport authorities need to collaborate to establish standardized guidelines for information exchange among sensor-equipped automobiles, as well as emergency response providers. By improving these types of specifications, government

bodies can generate a reliable and seamless connection system that helps instant vehicle crash reporting, decreases reaction periods, and possibly protects life.

Another important policy recommendation is the determination of financial resources to assist in the investigation of superior sensor fusion approaches. Authorities need to motivate relationships among the private and public sectors to promote technical development, making these platforms more cost-effective and available. Furthermore, concerns associated with cybersecurity and information privacy need to be resolved via legal guidelines, which protect user information while permitting efficient and safe data sharing among stakeholders. Motivating the incorporation of vehicle crash recognition platforms with emergency providers, for example, traffic control centers, EMS, and police stations, can additionally improve their performance, eventually decreasing death rates and enhancing highway safety.

To facilitate a powerful connection between emergency services and sensor-equipped automobiles, an incorporated information exchange system needs to be established. This particular system can work via a centralized cloud-based platform that aggregates live information through in-vehicle IoT sensors as well as instantly forwards it to the appropriate specialists, for example, police departments, traffic control units, and EMS. Developing a local or nationwide smart transportation system would help with connection capabilities, data governance, and system maintenance. Furthermore, the utilization of the current Intelligent Transportation System (ITS) infrastructure will enhance automobile accident recognition, making the entire crash management platform much more efficient and responsive.

5.2. Limitations

In spite of the developments in sensor systems, various restrictions prevent the extensive adoption of IoT-based vehicle crash recognition platforms. The most considerable issue is actually the dependability of sensors in various weather circumstances. Variables, for example, severe temperature, highway vibrations, as well as hardware failures, may impact sensor precision, resulting in wrong alerts or an inability to recognize vehicle crashes. Furthermore, information transmission delays in distant or low-network coverage regions may reduce the live functions of these types of platforms, making them less powerful in specific areas. Another technical restriction is actually the dependence on constant repair, which may be complicated for automobiles working for long periods.

Cost is actually one of the most important obstacles to the execution of multi-sensor platforms, particularly in low-income areas. The incorporation of professional detectors, information processing devices, and connection modules may be costly, making it challenging for individual users and small businesses to adopt these types of systems. In addition, concerns about the storage of data, information collection, and sharing pose substantial legal and ethical challenges. Without strong protection measures, there is actually a possibility of unauthorized access to sensitive automobile and motorist information, increasing worries regarding monitoring and improper use. To overcome these kinds of restrictions, the upcoming study must concentrate on establishing the most affordable sensor solutions, enhancing network infrastructure, as well as applying strict information safety rules to ensure both user privacy and efficiency.

6. Conclusions and Future Work

The IoT-based accident detection platform, which combines various detectors, is actually an innovative solution that utilizes it to make real-time notifications to the suitable authorities, thereby protecting life, and lastly, enhancing road safety. This particular study carried out an extensive evaluation of sensor-based accident recognition platforms by examining 101 peer-reviewed articles published in the field between 2008 and 2025. The innovative contribution of this research lies in the advancement of an organized classi-

fication tree, which categorizes innovative strategies into specific groups depending on recognition techniques, sensor types, and incorporation frameworks. Contrary to previous fragmented research, this specific evaluation provides a consolidated perspective that assists in recognizing scientific gaps, for example, the requirement for standardization across systems, communication methods, and reliable multi-sensor integration. The results of all the comparison evaluations discovered that a wide variety of detectors, such as GPS (53 studies), accelerometers (36 studies), vibration sensors (28 studies), and GSM modules (42 studies), carry out a significant function in real-time accident recognition and automobile safety platforms. The growing dependence on sensor systems is evident in their performance, which enhances reaction times, reduces fatalities, and improves highway safety. Nevertheless, in spite of these developments, specific restrictions, such as sensor precision, weather conditions, and information transmission issues, prevent the full potential of these techniques from being realized. One of the key findings of this particular research is that incorporating multiple sensors substantially improves the precision and dependability of crash recognition systems. While GSM and GPS sensors help area monitoring as well as urgent communication, image processing and accelerometer approaches lead to discovering the level of intensity and automobile orientation. Nevertheless, dependence on a single detector usually results in issues, for example, a wrong alarm or failure to identify particular crash circumstances. This highlights the significance of choosing hybrid sensor fusion strategies to enhance accuracy, guaranteeing that accident recognition techniques can perform suitably in different circumstances. Furthermore, power and cost usage continues to be the main obstacles to widespread utilization. Several advanced sensor systems, such as MEMS and LiDAR, offer high accuracy but come with significant costs, which limits their adoption in commercial vehicles. Furthermore, constant sensor utilization requires significant amounts of energy, increasing worries regarding battery life and energy performance, particularly in IoT-enabled intelligent automobiles. Upcoming studies must concentrate on establishing cost-effective, low-power detectors that maintain substantial efficiency while guaranteeing low cost and durability in large-scale deployments. In spite of these kinds of issues, the possibility of improving crash recognition platforms remains substantial. Developments in ML, AI, and the IoT promise possibilities to further improve these kinds of platforms. AI-driven information evaluation is able to assist in forecasting vehicle crashes depending on sensor inputs, though an IoT connection can easily assist in real-time connection with EMS. By combining these kinds of systems with current sensor networks, upcoming platforms are able to improve predictive abilities, decrease reaction periods, as well as play a role in proactive vehicle crash avoidance rather than just post-crash recognition. Although existing platforms have created significant improvements, additional study and advancement are generally essential to overcome the current restrictions associated with power performance, cost, and information precision. The upcoming study should focus on enhancing IoT-based vehicle crash recognition through superior sensor fusion, AI-driven information evaluation, and seamless connection with EMS using new technologies. Strengthening cybersecurity as well as combining it with autonomous automobile devices can improve live responsiveness, scalability, and dependability for a more secure transport sector. By addressing these kinds of issues, experts and planners will be able to establish next-generation safety devices, which are very dependable, widely available, and lastly, able to protect life on a worldwide level.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
AV	Autonomous Vehicle
ADAS	Advanced Driver Assistance Systems
CCTV	Closed-Circuit Television
CART	Classification and Regression Trees
CNN	Convolutional Neural Network
CU	Control Unit
DL	Deep Learning
EMS	Emergency Medical Service
FAR	False Alarm Rate
GPS	Global Positioning System
GPRS	General Packet Radio Service
GMM	Gaussian Mixture Model
GSM	Global System for Mobile Communications
IOT	Internet of Things
IR	Infrared
IoV	Internet of Vehicles
ITS	Intelligent Transportation System
LoRa	Low-range radio
ML	Machine Learning
MEMS	Micro-Electromechanical system
OBU	On-Board Unit
OBD	On-Board Diagnostics
RFID	Radio Frequency Identification
SMS	Short Message Service
SVM	Support Vector Machine
VANET	Vehicular Ad hoc Network
V2I	Vehicle-to-Infrastructure
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-everything

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